Small Scale, Big Decisions: Evaluating Decision Reliability in Multilingual LLM Pretraining

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Abstract

Design choices for training large multilingual language models are typically guided by results from ablations performed using smaller proxy models. However, such results are only useful if evaluations of small-scale models reliably predict a decision's effects in larger-scale models. In this work, we study this question for multilingual LLMs, and pretrain a controlled ladder of 218 multilingual models, spanning scales and language mixtures and evaluate language modelling and downstream performance on 20 multilingual benchmark families. For each benchmark, we quantify how well the performance of smaller models predicts larger model performance, how early in training this correspondence emerges, and whether small-scale evaluations recover the relative effects of design choices at full scale. Our results show that the informativeness of small-scale multilingual results is set by multiple factors: proxy model size, the training tokens seen per-language, the item format, and the scoring rule. We convert these findings into concrete per-benchmark guidance to support the identification of evaluations that can lead to cheaper, better-informed multilingual model development.

1 Introduction

Modern language model development relies on smaller proxy models and intermediate training checkpoints to guide decisions about data mixtures, architectures, tokenizers, and other training choices. This practice rests on the assumption that measured performance of smaller training runs is predictive of the performance of larger training runs. This predictive property of an *evaluation configuration*, a benchmark task together with an answer format and a scoring rule, is its *decision reliability* (i.e., whether decisions at small scale would be the same as those taken for models of the reference scale).

While decision reliability has been previously studied (Magnusson et al., 2025), these works only conducted experiments in English settings. However, multilingual language modelling introduces a host of interacting (and potentially confounding) factors that influence conclusions of decision reliability. First, languages vary in training mixture allocation, tokenization cost, and scaling behaviour (Petrov et al., 2023; Ahia et al., 2023; He et al., 2025). Second, cross-lingual transfer and interference dynamics further couple decisions across languages, meaning that adding data for one language can help or hurt others depending on model capacity and language similarity (Chang et al., 2024; Longpre et al., 2026). Finally, many multilingual language–benchmark pairs are known to exhibit little measurable signal at small scales, with performance often remaining near chance or floor levels (Penedo et al., 2025; Kydlíček et al., 2024). These challenges suggest that decision reliability findings in monolingual settings cannot be transferred directly to multilingual settings.

In this work, we investigate decision reliability in multilingual settings. First, as decision reliability requires dependable benchmark measurements, we explore which minimal model sizes and training-token counts produce evaluation scores distinguishable from chance across a suite of common benchmarks (§4). Second, we investigate at which proxy model sizes and per-language training token counts the ranking of proxy models agrees with the ranking of fully trained target models (§5). Finally, we identify which statistics computed from the proxy runs alone are predictive of this decision reliability, such that a developer can estimate how much to trust a proxy evaluation configuration without needing to train the target model (§6).

To support these investigations, we pretrain a controlled multilingual language scaling ladder, spanning model sizes, language mixtures, and architecture design axes. We evaluate 218 models on up to 1,488 benchmark-language tasks and perform an analysis of decision reliability. We assess

ranking agreement between proxies and the reference across model size and language training tokens, and search for cheap surrogates for ranking agreement across multilingual regimes. We find that the per-language training tokens seen, expected answer format, and scoring strategy impact decision reliability more than changes to proxy model size across the scales we test. Reformatting questions as cloze items makes Belebele, Global-MMLU and INCLUDE scores measurable where accuracy when using their original letter formats stays at chance. Once above chance, task accuracy scales predictably across model size and training checkpoints, and log-likelihood-based scoring on benchmark gold answers improves ranking agreement. However, statistics computed from the proxy runs alone, used to predict ranking agreement in monolingual settings, do not transfer to our multilingual ladder. We convert the measurements into the per-benchmark guide released with this paper. In summary, our contributions are:

- An **open controlled multilingual scaling ladder** together with evaluation measurements.[1]
- A **guide to evaluation configuration** for multilingual ablations. For each benchmark, we identify the per-language training-token thresholds at which training configurations pass the above-chance filter and the counts at which we observe decision reliability.[2]
- An **analysis of general trends in decision reliability** to inform multilingual ablation design, including recommendations on task filtering, reformatting, and scoring.

2 Related Work

Benchmark Reliability. Recent work shows that benchmarks differ substantially in how reliably they inform language model development. Heineman et al. (2025) introduce a signal-to-noise ratio (SNR) framework where signal measures a benchmark's ability to separate models and noise captures score variability across checkpoints or stochastic runs. They show that high-SNR benchmarks better preserve rankings and that low-noise benchmarks reduce scaling-law prediction error. Complementary work studies benchmark quality through ranking consistency, discriminability, and capability-alignment deviations (Qian et al., 2026), while item response theory (IRT) methods show that benchmark items differ widely in how well they discriminate between models, and that small, informative subsets can often recover the ranking of the full benchmark (Zhou et al., 2026; Li et al., 2026). Scaling-law studies further suggest that only some evaluations provide predictable development signal (Chen et al., 2025; Schaeffer et al., 2026). Together, these papers motivate our focus on identifying which multilingual measurements remain reliable across model sizes and nearby pretraining checkpoints, rather than treating all benchmark averages as equally meaningful.

Multilingual Evaluation. Multilingual benchmarks inherit reliability problems from their English counterparts and add new ones. Most popular benchmarks are machine-translated from English, which introduces translation artifacts, cultural mismatch, and label noise that distort cross-lingual rankings (Plaza et al., 2025; Singh et al., 2025; Romanou et al., 2025). Tokenizer compression varies by an order of magnitude across languages, inflating both training and evaluation cost for non-Latin scripts and morphologically rich languages (Ahia et al., 2023; Petrov et al., 2023). Cross-lingual transfer further couples decisions across languages, since adding data for one language can help or hurt others depending on model capacity and language similarity (Chang et al., 2024; Longpre et al., 2026). These factors suggest that estimates from English alone do not transfer directly across languages, and motivate a multilingual-specific framework built on controlled pretraining runs.

3 Experimental Setup

We study the problem setting in which a developer chooses between different design choices in language model pretraining. We consider choices related to model architecture, optimization, and multilingual data composition, and study them across different model sizes and training checkpoints. A *training configuration* is a model size with one value of each design choice of Section 3.1; a *run* comprises one configuration and one random seed; *model variants* are training configurations that differ in a design choice, and a *variant pair* is a pair of them. We ask whether the ranking of model variants at smaller sizes or earlier checkpoints is preserved at the largest model size or after full training. We describe the model ladder (§3.1) and the evaluation protocol used to rank models (§3.2).

[1] All artifacts will be publicly released upon publication.
[2] https://scaling-reliability.netlify.app

3.1 Model Ladder

We construct a controlled multilingual ladder of 218 models to measure ranking agreement across model size, training progress, and pretraining design choices. We train seven model sizes with 90M, 175M, 350M, 600M, 1B, 1.7B, 3B non-embedding parameters. We use the models between 90M to 1B parameters to predict performance at the 1.7B scale, and evaluate whether these findings generalize to the larger 3B scale. All models share the same training recipe, except for the design choices varied in each intervention. We use the Apertus architecture (Hernández-Cano et al., 2026), developed for multilingual language modeling, as our default configuration and perform controlled interventions to evaluate the impact of various design choices. Models are trained using the DCLM-Edu and FineWeb-2 datasets with a token budget proportional to model size and a warmup-stable-decay learning-rate schedule. Pretraining the full model ladder consumed 199,200 GPU hours. Appendix A.1 provides the complete training grid and hyperparameters.

Intervention axes. We consider six pretraining design choices:

- **Number of training languages.** We train models on $1, 2, 8, 15, 30, 50$ languages. The monolingual setting uses only English, while the multilingual settings allocate half the training tokens to English and half to the remaining languages.
- **Language selection scheme.** We compare two approaches for selecting the non-English training languages. Scheme A prioritizes languages with larger available corpora, whereas scheme B prioritizes diversity in scripts and language families. Appendix B lists the languages selected by each scheme at every language count.
- **Language sampling temperature.** We compare two sampling temperatures for allocating training tokens across non-English languages: *proportional sampling*, which reflects the relative size of each language's available corpus, and *flattened sampling*, which increases the representation of languages with smaller corpora.
- **Model depth.** We compare a deep with a shallow architecture. The shallow models have approximately twice the width-to-depth ratio while maintaining the same feed-forward multiplier and ratio of query to key-value attention heads.
- **Optimizer.** We use AdEMAMix by default (Pagliardini et al., 2025). For models trained on 8, 15, or 30 languages, we also evaluate Muon (Jordan et al., 2024), following the Moonlight recipe (Liu et al., 2025).
- **Activation.** xIELU is the default activation function (Huang and Schlag, 2025). For deep models trained on 8, 15, or 30 languages, we also evaluate SwiGLU (Shazeer, 2020).

3.2 Evaluation

We evaluate all models at aligned checkpoints throughout training to compare their rankings across model sizes and training progress. Specifically, we evaluate models at every 10% of their training budget, from 10% to 100%, with two additional checkpoints at 85% and 95% used to estimate checkpoint noise. At each checkpoint, we measure model performance using held-out text likelihood and downstream benchmarks. Appendix C provides the complete evaluation setup and implementation details.

Evaluation suite. We evaluate models using held-out text bits-per-byte (BPB) and 52 downstream benchmarks covering 50 languages and a range of capabilities, including reading comprehension, commonsense reasoning, factual knowledge, and grammatical acceptability (Table 6). BPB measures the negative log-likelihood of held-out text normalized by its UTF-8 byte count, with lower values indicating better performance. We include BPB as a reference to established loss-based scaling behaviour and a baseline for downstream benchmarks. By default, each model is evaluated only on languages included in its training mixture.

Answer formats & scoring protocols. The downstream benchmarks include three answer formats: multiple-choice letters, cloze, and completion. We evaluate these tasks using length-normalized accuracy and benchmark bits-per-byte (bBPB). Accuracy measures whether the model selects the correct answer, while bBPB measures the negative log-likelihood of the gold answer normalized by its UTF-8 byte length, with lower values indicating better performance. Unlike accuracy, bBPB does not have a chance-level baseline, allowing us to evaluate benchmarks even when model accuracy is near chance.

4 Measurability: When Do Benchmarks Become Measurable?

A score that is statistically indistinguishable from chance provides no reliable information about

which model is better. We therefore first assess the *measurability* of a training configuration on a task, which determines whether a score is statistically distinguishable from chance, as a prerequisite for examining decision reliability. For each training configuration, we compute the one-sided 95% Wilson lower confidence bound on its accuracy $\hat{p}$ over the task's n items. We say that a training configuration is *above chance* on the task when this bound exceeds $1/m_b$, where m_b is the number of answer choices; this test on one configuration's result is the *above-chance filter*. Since the bound depends on n, tasks with fewer items require a larger margin above chance.

This section examines, for each benchmark–language task, which training configurations score above chance. We say that a model-size passes the filter for a given benchmark–language task when at least half of the training configurations at that model size score above chance. The benchmark–language–model-size combinations that don't pass are excluded from later analyses. BPB and bBPB are exempt from the filter because they have no defined chance level. We report how this pass rate changes with model size and as a function of *target-language tokens*—the number of training tokens seen by the evaluated model on the given task's language (§4.1). We then relate it to the answer format (§4.2) and ask whether model performance on the filtered tasks scales predictably (§4.3).

4.1 Model Size and Token Count Thresholds

We analyse 938 benchmark–language tasks, listed in Table 6. For only 38% of tasks does every model size pass the above-chance filter; for 34% of tasks, no model up to 1.7B passes the filter (Figure 5, Appendix D.1). The tasks on which no model size passes the filter are predominantly from benchmarks requiring letter-format answers like Belebele, Global-MMLU and INCLUDE, for which the filter retains at most three language tasks at any model size. In contrast, on benchmarks scored on the answer text, such as HellaSwag (Okapi) and MultiBLiMP, training configurations score above chance in all trained languages even at our smallest size (90M). This finding suggests that evaluation format affects measurability, which we explore below (§4.2). Full results are provided in Figure 6.

Model size is not the only factor that determines whether a configuration scores significantly above chance on a task. We also ask how many target-language tokens a model needs to have seen for it to pass the filter on a given task. Across non-English benchmarks, training configurations with $\leq$100M target-language tokens score above chance on 42% of tasks, while the proportion reaches 64% with 10B–100B tokens. Accuracy therefore stays at chance on a third of tasks even after training on 10B tokens of the target language, the full Chinchilla budget of a 500M monolingual model. Some notable exceptions are HellaSwag (Okapi), MultiBLiMP and TokSuite, on which nearly all training configurations with fewer than 100M target-language tokens score above chance. The effect of target-language tokens also varies with model size, with 43% of training configurations at 90M parameters scoring above chance after 100M–1B target-language tokens, compared with 62% at 1.7B. These findings suggest that both the number of target-language tokens and model size influence measurability (Table 7). Full benchmark-level results are provided in Figure 8 (Appendix D.3).

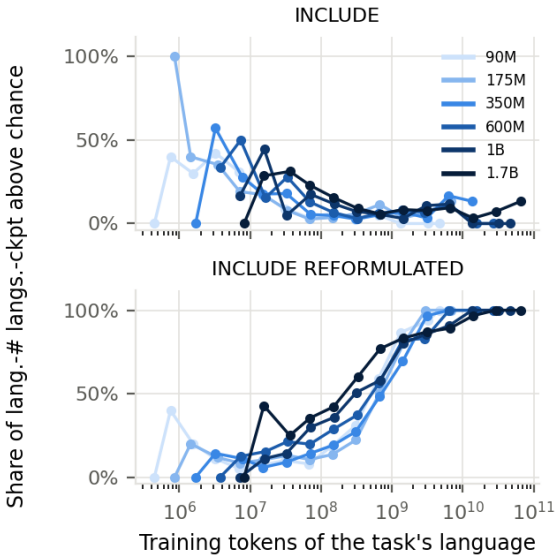


Figure 1: Original (top) and reformulated (bottom) INCLUDE results across training. Plots for all benchmarks are in Figure 8.

4.2 Answer Format and Reformulation

Multiple-choice benchmarks, where the answer is a letter, require mapping answer labels to their content, an ability that may only emerge at larger scales or with fine-tuning, regardless of whether the model knows the correct answer (Khatun and Brown, 2024; Madaan et al., 2024; Gu et al., 2025). To isolate the effect of task format, we reformulate these benchmarks to score the text of each answer option directly. We consider two

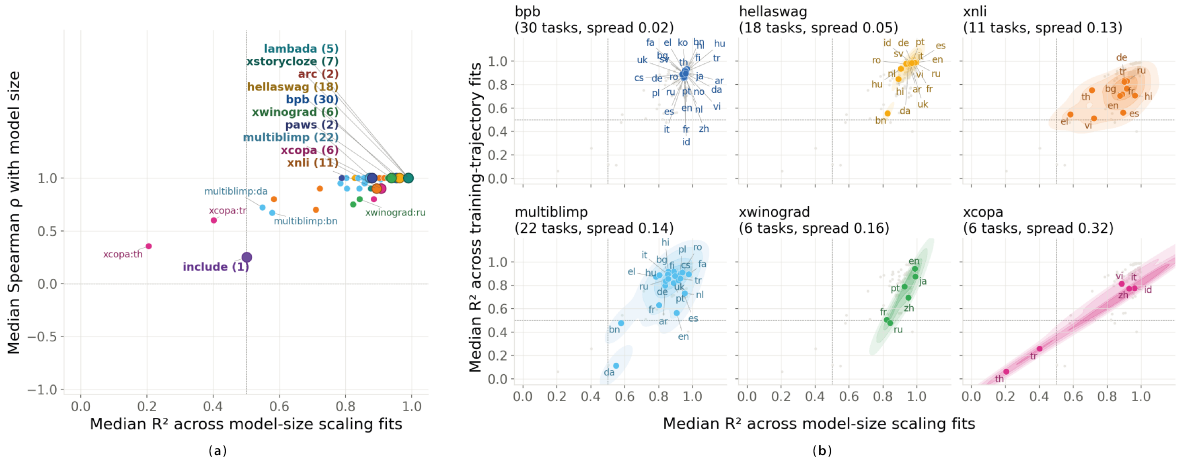


Figure 2: **Benchmark scaling predictability. (a)** Median R^2 versus Spearman ρ across model-size fits for each benchmark, showing scaling predictability and direction. **(b)** Median R^2 across model-size and training-trajectory fits for selected benchmarks, with each point representing a benchmark–language task. Detailed scaling curves across model sizes and training checkpoints are provided in Figures 9 and 10 (Appendix D.4)

approaches: *programmatic reformulation*, which removes the answer letters without changing the question or the options, and *model-based reformulation*, which uses `gemini-3.8-flash` (Google DeepMind, 2026) to rewrite each question and options as declarative statements, without revealing the correct answer.

These reformulations substantially increase the number of languages for which training configurations pass the above-chance filter. At the 1.7B size, programmatic reformulation increases this number from 5 to 86 of 105 languages for Belebele, from 1 to 35 of 37 for Global-MMLU, and from 3 to 31 of 43 for INCLUDE. Figure 1 illustrates this effect for INCLUDE. In the original letter format, accuracy rarely exceeds chance, while in the reformulated version, the share of training configurations that pass the filter rises as training progresses. Model-based reformulation yields further improvements: on Belebele, every model size passes the filter in all trained languages. Full results are in Figure 7.

4.3 Benchmark Scaling Behavior

Next, we examine how predictably performance scales along both dimensions. We assess scaling predictability by fitting task performance against log model size at the final checkpoint and against log training tokens across checkpoints, using R^2 to measure goodness of fit and Spearman's ρ to measure monotonicity and direction. Full fitting details are provided in Appendix D.4.

Figure 2a shows substantial variation in scaling predictability across benchmarks. Across 431 tasks retained by the above-chance filter, the median R^2 of the model-size fits is 0.87, reaching 1.00 for the BPB across 50 languages (Appendix D.4). For the letter-format benchmarks (Global-MMLU, INCLUDE and Belebele), too few model sizes pass the filter for fitting, but the model-size fits of their reformulated versions achieve a median R^2 of 0.84–0.92 (Appendix D.4, with the curves in Figures 9 and 10). A high R^2, though, does not necessarily imply improvement across scales. For example, TruthfulQA has a median model size-fit R^2 of 0.97 but $\rho = -0.75$, indicating a strong and consistent decrease in predictability with model size.

Figure 2b compares scaling predictability across model size and training checkpoints. Among accuracy and BPB tasks with valid fits along both dimensions, 74% (311 tasks) achieve $R^2 \geq 0.5$ on both. Notably, no accuracy or BPB task exhibits predictable scaling across training checkpoints without also scaling predictably across model sizes, although the reverse is not always true. The same pattern holds for monolingual English models (Appendix D.4). We therefore recommend, within the scales we test, spending ablation compute on additional model sizes rather than on a longer run of one size: the model-size fits give a consistent trend on every task the training fits do, and on more.

To make these results practically useful, Table 7 (Appendix D.3) provides, per benchmark and an-

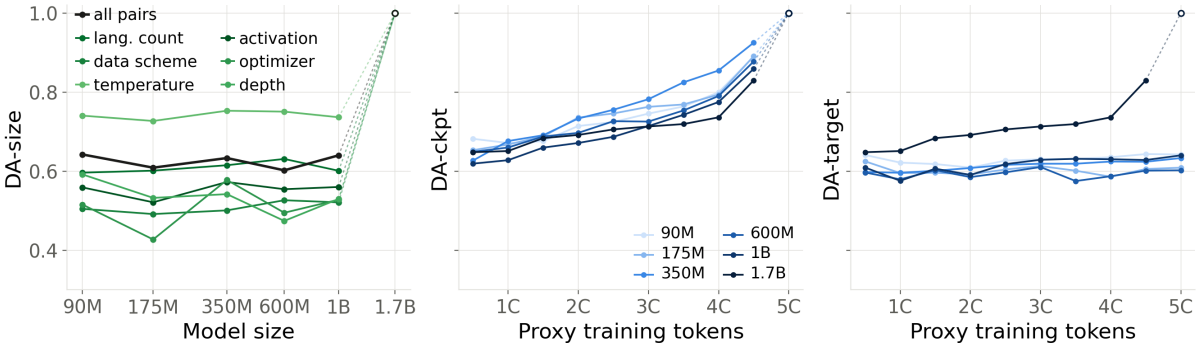


Figure 3: Decision accuracy of proxy models for the 1.7B reference, over the benchmark–language tasks (accuracy and bBPB) whose median DA in the scale or within-model training agreement setting is at least 0.66. **(a) Scale agreement**: DA of the final checkpoint of each model size against the final 1.7B ranking, pooled over decisions, one line per design axis and a black line over all pairs. **(b) Within-model training agreement**: DA of each intermediate checkpoint relative to the final checkpoint of the same model size, averaged over tasks. **(c) Target agreement**: DA of each intermediate checkpoint relative to the final 1.7B reference model, measuring how early partially trained and smaller models recover the decision at the reference size.

swer format, the *above-chance thresholds*: the target-language token counts at which at least 50% and 90% of the evaluations pass the filter.

Summary. A large share of the suite does not perform measurably on these benchmarks. In 34% of the 938 analysed benchmark–language tasks, no model size up to 1.7B passes the above-chance filter; these tasks are predominantly those scored using multiple-choice answer letters. Whether a training configuration scores above chance on a task depends on both the target-language tokens it has seen and its model size. On the tasks the filter retains, accuracy scales predictably with model size and with training tokens.

5 Agreement: How Early Can We Trust Proxy Evaluations?

Building on the results from the previous section, we investigate whether evaluations at smaller model scales or earlier training stages can reliably inform decisions made at full scale. We study ranking agreement by comparing the relative ordering of model variants at a cheaper proxy point (model size and training checkpoint) with a reference point (the final checkpoint at the same or largest model size). We examine three settings: *Within-model training agreement*, which compares intermediate checkpoints with the final checkpoint at the same model size (§5.1); *Scale agreement*, which compares fully trained smaller models with the fully trained largest model (§5.2); and *Target agreement*, which evaluates whether intermediate checkpoints across model sizes recover the rankings of the fully trained largest model (§5.3). These settings are summarised in Table 8, with formal definitions in Appendix E.1. We additionally examine the sensitivity of ranking agreement to near-tied model scores, evaluation scoring protocols, and per-language training-token budgets, as detailed in the appendix.

We quantify ranking agreement using decision accuracy (DA; Magnusson et al., 2025), defined as the fraction of pairwise model preferences preserved between the proxy and reference points. We compute DA only for ranking comparisons involving at least three variant pairs, since comparisons between only two pairs yield binary agreement scores. We compare model variants across the intervention dimensions defined in Section 3.1, using model sizes ranging from 90M to 1.7B parameters. We focus on accuracy and bBPB tasks whose median DA across proxy points reaches at least 0.66 in either model-size or training-checkpoint DA. This heuristic threshold selects tasks with higher observed ranking agreement.

5.1 Within-Model Training Agreement

Figure 3b shows the mean DA across benchmark–language tasks and model-variant comparisons at each training checkpoint, using the final checkpoint at the same model size as the reference. Ranking agreement generally increases with training progress across all model sizes, rising from approximately 62–68% at early checkpoints to 83–93% in the later stages of training. Nevertheless, disagreement persists even near the end of training, indicating that intermediate evaluations do not always reproduce final model-selection decisions. Al-

though 350M models generally exhibit the highest agreement, we observe no consistent relationship between model size and ranking agreement.

5.2 Scale Agreement

Figure 3a shows the mean DA between fully trained models at different scales and the fully trained 1.7B reference, across model-variant comparisons grouped by intervention type. Overall, agreement remains modest (59–61% across all pairs) and shows no consistent improvement with increasing model size. However, agreement varies substantially across intervention types, with sampling temperature consistently exhibiting higher agreement (72–75%) than other design choices. These results suggest that the reliability of cross-scale decisions depends more on the type of model intervention than on the size of the proxy model alone.

5.3 Target Agreement

Figure 3c shows the mean DA between intermediate checkpoints across model sizes and the target (the fully trained 1.7B reference model), measuring the combined effects of model scale and training progress. For smaller models, agreement remains modest (approximately 58–64%) and shows little systematic improvement with additional training. In contrast, agreement at 1.7B increases steadily throughout training, consistent with the within-model training agreement observed in Figure 3b. This contrast indicates that improvements in ranking agreement during training do not necessarily transfer across model scales, limiting the ability of smaller, partially trained models to predict decisions at the reference size. To test whether reliable cross-scale ranking agreement emerges at larger proxy sizes, we extend the model ladder to 3B for a subset of configurations. Even at 1.7B, benchmark accuracy shows low agreement with the 3B reference, with DA of 0.51–0.52 (Appendix E.6, Figure 21). This indicates that increasing proxy model size to 1.7B is still not sufficient to recover reliable rankings at the reference size from multilingual benchmark accuracy.

5.4 Sensitivity Analysis

We further examine the sensitivity of ranking agreement to near-tied model scores (i.e., model variants with very close rankings), evaluation scoring protocols, and per-language training-token budgets. Accounting for near-tied scores as expected increases DA without substantially changing the main trends, while BPB-based evaluations generally preserve rankings more reliably than benchmark accuracy, particularly at smaller scales and earlier training stages. These additional analyses are presented in Appendices E.2, E.3, and E.4.

Summary. On accuracy-scored benchmarks, the share of variant pairs that a proxy point orders the same way as the final 1.7B checkpoint is close to pairwise chance: 0.51-0.56 at every proxy size. On per-language BPB, the share is higher, ranging from 0.74 to 0.94 over the same proxies. Seed-to-seed noise in the reference models accounts for part of the disagreement. On the design-variant pairs that the 1.7B reference model separates by more than two seed standard deviations, the share ordered correctly is 0.69 to 0.73.

6 Anticipation: Can We Predict Ranking Agreement?

Computing DA requires access to both proxy and reference evaluations, limiting its utility for guiding development decisions before larger models are trained. Prior work has proposed several surrogate metrics, most notably signal-to-noise ratio (SNR; Heineman et al., 2025), which compares the separation between model variants to the variability of their evaluation scores. These methods have primarily been studied in English settings and at scales where downstream benchmark signal is already measurable. In this analysis section, we investigate whether existing surrogate metrics are associated with decision accuracy for multilingual scaling and whether they provide useful signal at the smallest model scale.

Specifically, we test whether statistics computed from proxy evaluations alone can serve as surrogates for DA. For each benchmark–language task and proxy size, we compute each surrogate without using the reference model. We then measure its Spearman correlation (ρ) with DA across benchmark–language tasks at each proxy size. Figure 4 reports these correlations across proxy sizes for tasks passing the above-chance threshold. We compare scaling-fit R^2, SNR variants (Heineman et al., 2025) and their signal/noise components, margin above chance, and early-checkpoint ranking agreement. These capture, respectively, how well scores follow a log-linear scaling trend, the separation and variability of model scores, performance relative to chance, and ranking agreement between early and final checkpoints at the same size. Formal definitions are provided in Appendix F.1.

We find that no single surrogate consistently predicts DA across multilingual settings (Figure 4).

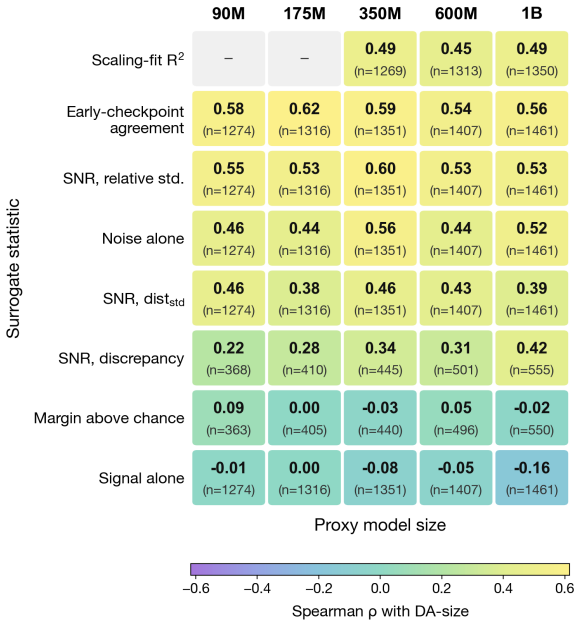


Figure 4: Spearman rank correlation (ρ) between each surrogate and DA for the 1.7B model across benchmark–language tasks. Cells report the ρ and sample size *(n)*.

Early-checkpoint ranking agreement ($\rho = 0.54$–0.62) and the relative-standard-deviation SNR ($\rho = 0.53$–0.60) show the strongest overall associations, followed by scaling-fit R^2 ($\rho = 0.45$–0.49). In contrast, signal alone and margin above chance show little association with DA. For SNR-based surrogates, however, performance depends strongly on how noise is defined, with some noise estimates yielding substantially stronger associations than others. These findings suggest that benchmarks whose rankings stabilize early in training and whose scores exhibit less evaluation noise are also more likely to preserve the ranking of model variants across scales. Stronger associations emerge for individual benchmarks, up to $\rho = 0.93$ for HellaSwag, demonstrating that predictability is benchmark-dependent.

We next examine how these associations change with model size. Moderate correlations are already present at small scales and do not consistently improve as models grow. Early-checkpoint agreement reaches $\rho = 0.62$ at 175M, while relative-standard-deviation noise reaches $\rho = 0.56$ at 350M. Scaling-fit R^2 becomes available from 350M and remains relatively stable ($\rho = 0.45$–0.49) up to 1B. Overall, surrogate signal emerges at small scales but does not consistently strengthen with model size. This suggests that reliable evaluation at smaller scales does not necessarily imply consistent model rankings at larger scales. For example, at 600M, per-language BPB shows low evaluation noise (checkpoint noise of 0.4% against 1.3% of the mean score), strong scaling fits (median R^2 of 1.00), and clear score differences between variants (median relative-standard-deviation SNR of 8.7 against 2.0 for benchmark tasks), yet DA against the 1.7B reference remains at 0.70–0.74 throughout training. This highlights the distinction between reliably measuring performance differences at the proxy scale and preserving model rankings across scales.

Summary. No metrics computed from proxy evaluations predicts DA consistently across benchmarks or proxy sizes. The two metrics with the strongest association are early-checkpoint ranking agreement, the DA between a run's 10% checkpoint and final checkpoint ($\rho = 0.54$–0.62), and the relative-standard-deviation SNR, the spread of the variants' final scores divided by checkpoint noise ($\rho = 0.53$–0.60).

7 Conclusion

We study when small-scale multilingual evaluation supports reliable pretraining decisions using a controlled ladder of 218 models. Our findings challenge the assumption that ablations at smaller scales reliably identify the best choices for larger multilingual models. Measurable scores, smooth scaling curves, and stable proxy rankings do not guarantee cross-scale agreement, and increasing proxy size does not consistently resolve this problem within our ladder. Reformulating tasks can improve evaluations' decision reliability, but reliability remains conditional on the benchmark, language, and intervention axis.

These results raise a broader question: how reliably do current model-development practices select the best pretraining choices? If proxy experiments are noisy or favour choices that become inferior at larger scales, they may lead developers to discard promising alternatives and limit the performance of the resulting models. Our study does not establish how often this occurs in practice or quantify its impact, but suggests the issue may be more prevalent than currently believed. Future work should therefore investigate two questions: how much performance have current models lost through unreliable proxy decisions, and how can we better predict the effects of pretraining choices at the reference scale? We release our model ladder, evaluation measurements, and per-benchmark guide to help assess these conditions.

Limitations

Compute constraints limit the model sizes, language mixtures, and pretraining interventions we test, with comparisons at 3B covering only a subset of configurations. However, the controlled model ladder allows us to study these factors systematically within the available budget. Within this ladder, most configurations use a single seed, leaving our estimates of decision reliability sensitive to noise in the reference rankings. Moreover, the above-chance threshold changes the set of tasks included at each scale and complicates comparison, but it prevents near-chance variation from driving the analysis, and we report fixed-population comparisons.

Our analysis is also limited by multilingual benchmark coverage and validity. Although our training mixtures include up to 50 languages and we made an effort to broaden evaluation coverage, few benchmarks are available for many low- and mid-resource languages. Even where benchmarks exist, web-derived training data may not represent the full range of varieties, domains, and speaker communities within each language. We do not check for benchmark contamination, and model-based reformulation may introduce errors or alter item meaning. To minimize the latter, we manually review the generations in six languages and iterate on the prompt.

Ethical Considerations

Training and evaluating the ladder required substantial compute. Controlled pretraining experiments are necessary to separate the effects of design choices, although their environmental cost remains a concern. We release the models and measurements to support reuse and reduce the need to repeat these experiments.

Our models may inherit potential risks from their web-sourced training corpora. We rely on the source datasets' filtering and apply no additional filtering for personal information or offensive content. We also do not perform any safety tuning, so the models may reproduce sensitive information or generate harmful, biased, or misleading content. We release them for research on pretraining and evaluation, with accompanying documentation of intended uses.

Finally, uneven language coverage presents a separate risk: evaluation-driven decisions may favour better-represented languages and communities. Our efforts to broaden coverage were constrained by available data and benchmarks. We therefore document the tested languages, limit our claims to the observed settings, and call for benchmarks in underrepresented languages to support broader assessment.

AI Use Statement

We used a language model to reformulate benchmark questions and answer options as cloze items, as described in the methodology. Generative AI tools were also used for code review, preliminary result visualization, spelling and grammar correction, formatting, and minor language editing of the manuscript. They were not used to conceive the methodology, design the experiments, interpret the results, or draft the scientific analysis. All AI-assisted content was reviewed by the authors, who take responsibility for the final manuscript and its scientific claims.

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A Model Ladder

We construct a controlled multilingual ladder of 218 models to measure ranking agreement across model size, training progress, and training design interventions. We train seven model sizes, measured in non-embedding parameters: 90M, 175M, 350M, 600M, 1B, 1.7B, 3B. The models from 90M to 1B are the proxies, 1.7B is the reference, and 3B tests whether the conclusions hold one size above. All runs share the same training recipe described below, except for the factor varied by each intervention. This appendix describes the shared configuration (§A.1), the intervention axes (§A.2) and the training grid (§A.3). Appendices B and C.2 describe the data mixtures and the evaluation protocol. All models will be released under the Apache License 2.0.

A.1 Shared Configuration

Parameter count. All sizes are non-embedding parameters, that is, the transformer blocks alone, without the tied token embedding and output projection. This follows the ladder convention of Bhagia et al. (2025), which matters because of the 131,072-token vocabulary. The tied embedding matrix alone has $d_{\text{model}} \times V$ parameters, 101M at 90M and 302M at 1.7B, so counting it would substantially compress the spacing of the smallest models sizes.

Training budget. Each model receives a token budget proportional to its exact non-embedding parameter count N: $D(N) = 100N$. This corresponds to five times the $20N$ Chinchilla token-budget convention, following the fixed-multiple ladder design of Magnusson et al. (2025).

Training corpora. The English data comes from DCLM-Edu (Ben Allal et al., 2025), filtered from DCLM (Li et al., 2024), and the data of the other languages from the quality-filtered FineWeb-2 collection (Penedo et al., 2025; Messmer et al., 2025).

Architecture. All models are decoder-only transformers that follow the Apertus pretraining recipe (Hernández-Cano et al., 2026). We use grouped-query attention (Ainslie et al., 2023) with four query heads per key-value head and a head dimension of 64. We use rotary position embeddings (Su et al., 2024) with base 500,000 and Llama-3-style scaling factor 8, RMSNorm (Zhang and Sennrich, 2019), and QK-layernorm (Henry et al., 2020; Dehghani et al., 2023). The models have no bias terms and no dropout. Input and output embeddings are tied over the 131,072-token Apertus tokenizer.[3] The sequence length is 4,096.

Learning rate. We use a warmup-stable-decay learning-rate schedule (Hägele et al., 2024). The global batch size is 504 sequences at every size except 90M and 175M, which use 84 and 168. The peak learning rate follows the compute-based fit of DeepSeek-AI et al. (2024) at the run's budget, with compute approximated as $6ND$, and the initialization standard deviation scales with the model width (Yang et al., 2021; Hu et al., 2024). Table 2 lists the hyperparameters of every size.

A.2 Intervention Axes

Number of training languages (K). We train with $K \in \{1, 2, 8, 15, 30, 50\}$ languages. The $K = 1$ setting is English only. Multilingual settings allocate half the training tokens to English and half to the other $K - 1$ languages.

Language selection scheme (M). We compare two rules for choosing the non-English languages. Scheme A ranks languages by available corpus size, whereas scheme B prioritizes diversity in script and language family. At $K = 2$ the schemes vary the second language: Russian (A), Chinese (B), or Spanish (C). At $K = 1$ they vary the English corpus: DCLM-edu (A), the same DCLM pipeline without its educational-quality filter (B), or FineWeb (C). Appendix B lists the languages selected by each scheme at every value of K.

Sampling temperature (T). Within the non-English half of the training mixture, language i has a sampling probability proportional to $q_i \propto p_i^{1/T}$, where p_i is the language's estimated proportion of the available corpus. We use $T = 1$ for proportional sampling and $T = 3$ for flattened sampling. The $T = 3$ runs use the scheme-A lists at $K \in \{15, 30, 50\}$, with both architectures at $K = 50$ and the deep architecture only at $K = 15$ and $K = 30$.

Depth (D). We compare a deep with a shallow-wide architecture at each size up to 1.7B. The deep models keep a width-to-depth ratio $d_{\text{model}}/n_{\text{layers}}$ of 64 from 175M to 1B and a feed-forward multiplier of 4. The ratio is 51 at 90M, 77 at 1.7B and 78 at 3B. The shallow models roughly double

[3] https://huggingface.co/swiss-ai/Apertus-70B-2509

Symbol	Meaning	Values
N	Non-embedding parameter count	90M, 175M, 350M, 600M, 1B, 1.7B, 3B
A	Architecture depth	Deep or shallow
O	Optimizer	AdEMAMix or Muon
A_f	Activation function	xIELU or SwiGLU
K	Number of training languages	1, 2, 8, 15, 30, 50
M	Language-selection scheme	A, B, C
T	Sampling temperature	1 or 3
S	Random seed	1904 (default), 313, 64
f	Training progress	0.1, 0.2, . . . , 1.0, plus 0.85 and 0.95

Table 1: **Notation for the model ladder.** Training progress f is the share of a run's token budget. Every run trains on $5 \times 20N$ tokens, so $f = 0.2$ is one Chinchilla multiple (1C) and $f = 1$ is 5C.

Ladder	Size	Layers	d_{model}	$d_{\text{model}}/n_{\text{layers}}$	FFN	FFN/d_{model}	Heads	N (M)	D (B)	Batch	Steps	LR (10^{-3})
deep	90M	15	768	51	3,072	4.00	12/3	92.9	9.3	84	27,000	1.43
deep	175M	16	1,024	64	4,096	4.00	16/4	176.2	17.6	168	25,620	1.22
deep	350M	20	1,280	64	5,120	4.00	20/5	344.1	34.4	504	16,660	1.03
deep	600M	24	1,536	64	6,144	4.00	24/6	594.5	59.5	504	28,800	0.90
deep	1B	28	1,792	64	7,168	4.00	28/7	944.1	94.4	504	45,720	0.80
deep	1.7B	30	2,304	77	9,216	4.00	36/9	1,672.2	167.2	504	81,000	0.69
deep	3B	36	2,816	78	11,264	4.00	44/11	2,997.5	299.7	504	145,200	0.60
deep + SwiGLU	90M	15	768	51	2,048	2.67	12/3	92.9	9.3	84	27,000	1.43
deep + SwiGLU	175M	16	1,024	64	2,752	2.69	16/4	177.2	17.7	168	25,740	1.21
deep + SwiGLU	350M	20	1,280	64	3,392	2.65	20/5	342.4	34.2	504	16,580	1.03
deep + SwiGLU	600M	24	1,536	64	4,096	2.67	24/6	594.5	59.5	504	28,800	0.90
deep + SwiGLU	1B	28	1,792	64	4,800	2.68	28/7	947.3	94.7	504	45,880	0.80
deep + SwiGLU	1.7B	30	2,304	77	6,144	2.67	36/9	1,672.2	167.2	504	81,000	0.69
shallow	90M	8	1,024	128	4,096	4.00	16/4	88.1	8.8	84	25,560	1.45
shallow	175M	10	1,280	128	5,120	4.00	20/5	172.0	17.2	168	25,020	1.22
shallow	350M	14	1,536	110	6,144	4.00	24/6	346.8	34.7	504	16,800	1.03
shallow	600M	14	2,048	146	8,192	4.00	32/8	616.6	61.7	504	29,860	0.89
shallow	1B	17	2,304	136	9,216	4.00	36/9	947.6	94.8	504	45,920	0.80
shallow	1.7B	20	2,816	141	11,264	4.00	44/11	1,665.3	166.5	504	80,640	0.69

Table 2: **Pretraining hyperparameters per ladder and model size.** N counts non-embedding parameters. The tied embedding adds $d_{\text{model}} \times 131{,}072$. $D = 100N$ is the token budget. Heads gives the query and key-value head counts (grouped-query attention, four query heads per key-value head, head dimension 64). FFN is the feed-forward hidden dimension. The SwiGLU ladder keeps the deep shape and narrows the FFN so that its three projections match the deep non-embedding count. The Muon ladder has the same values as the deep one. Batch is the global batch in sequences of 4,096 tokens. The two smallest sizes train at a smaller batch and take proportionally more steps for the same D. LR is the peak learning rate at the run's own budget.

the width-to-depth ratio (110–146 against 51–77) while preserving the feed-forward multiplier and the ratio of query heads to key-value heads. We restricted d_{model} to multiples of 256. Their non-embedding counts differ from the corresponding deep models by 0.4% to 5.2%.

Optimizer (O). The default optimizer is AdEMAMix (Pagliardini et al., 2025) ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\beta_3 = 0.9999$, $\alpha = 8$), with weight decay 0.1 and gradient clipping at 0.1. At $K \in \{8, 15, 30\}$ we also train the deep architecture with Muon (Jordan et al., 2024), following the Moonlight recipe (Liu et al., 2025). Muon updates the hidden weight matrices and rescales the orthogonalized update U of an $m \times n$ matrix to $\tilde{U} = 0.2\sqrt{\max(m, n)}\, U$, so that its size matches that of AdamW and the same peak learning rate and weight decay apply. Embeddings, the output layer and norms keep AdamW.

Activation (A). xIELU is the default activation function (Huang and Schlag, 2025). At $K \in \{8, 15, 30\}$, we also train the deep architecture with SwiGLU activation (Shazeer, 2020), with the feed-forward width reduced to keep the parameter count constant. This width shrinks to about $2.67\, d_{\text{model}}$, so that the three SwiGLU projections match the deep non-embedding count (within 0.6%).

Table 3 lists every data build and ladder of the training grid with its number of runs.

A.3 Training Grid

Grid. A run is defined by its size, its number of training languages K, its ladder, its data build

Build	Data	M	T	Ladder	K	Sizes	Runs	Repl.
A	resource-ranked lists	A	1	deep	1, 2, 8, 15, 30, 50	90M–1.7B and 3B at $K \in \{8, 15, 30\}$	57	18
A	resource-ranked lists	A	1	shallow	1, 2, 8, 15, 30, 50	90M–1.7B	36	–
A	resource-ranked lists	A	1	deep + SwiGLU	8, 15, 30	90M–1.7B	18	–
A	resource-ranked lists	A	1	deep + Muon	8, 15, 30	90M–1.7B	18	–
AT3	scheme-A lists, flattened	A	3	deep	15, 30, 50	90M–1.7B	18	–
AT3	scheme-A lists, flattened	A	3	shallow	50	90M–1.7B	6	–
B	diversity-first lists	B	1	deep	8, 15, 30	90M–1.7B and 3B at $K \in \{8, 15, 30\}$	23	2
B	diversity-first lists	B	1	shallow	8, 15, 30	90M–1.7B	18	–
ZH	Chinese as second language	B	1	deep	2	90M–1.7B	6	–
ES	Spanish as second language	C	1	deep	2	90M–1.7B	6	–
DCLMP	DCLM without the edu filter	B	1	deep	1	90M–1.7B	6	–
FWEB	FineWeb (crawls $\leq$ 2022)	C	1	deep	1	90M–1.7B	6	–
Total							218	20

Table 3: **Training grid.** One row per data build and model ladder. M is the language-selection scheme of the build at its language count K: A is the baseline recipe, and B and C are the alternatives at that K. T is the sampling temperature of the build. Runs counts the models of every seed, and Repl. counts the replicate seed runs among them.

and its seed. A ladder specifies the model recipe (architecture, activation, and optimiser), and a data build specifies the training grid data (Appendix B). The deep ladder on data build A is the baseline recipe. Table 3 lists every combination of data build and ladder that we train.

Seeds. Every configuration trains with seed 1904. A subset of the deep scheme-A configurations also trains two replicate seeds: 64 and 313 at 175M and 600M ($K \in \{1, 2, 50\}$), and 28 and 1797 at 1B ($K \in \{1, 2, 30\}$, and scheme B at $K = 30$). A seed changes both the initialization and the data order, so the replicates measure the combined effect of training randomness.

Checkpoints. Each run saves 20 evenly spaced checkpoints (40 at 1B, 60 at 1.7B and 3B), so every run contains the shared points $k/20$. We evaluate twelve of them. The ten tenths of training are shared by every size, so sizes can be compared at the same Chinchilla multiple (20% of a run is 1C and its end is 5C), and every analysis along a run reads them. The two extra checkpoints at 85% and 95% complete five evenly spaced points in the last 20% of training (80, 85, 90, 95 and 100%), the decay phase of the schedule, over which we estimate checkpoint noise.

Compute. Every checkpoint has a known (N, D), so results can also be read against training compute. We estimate FLOPs $= 6\,(N + d_{\text{model}}V)\,D$, where V is the vocabulary size. This omits the embedding lookup but counts the output projection, which nearly doubles the per-token cost of the smallest sizes. We train on nodes of four NVIDIA GH200 GPUs, using 3 to 21 nodes per run depending on size. Pretraining the full model ladder consumed 199,200 GPU hours.

B Data Mixtures

Each data build fixes three things: the list of training languages, the English corpus and the sampling temperature across languages. This appendix describes how we choose the language lists, how we mix the languages, how large each build is, and the validation set that every model shares.

Scheme A lists. We derive the language lists from the FineWeb-2 per-language byte counts and from the benchmarks available in the evaluation harness. *Scheme A* prioritises data availability. It ranks the FineWeb-2 subsets by the UTF-8 bytes of their training split, leaves out the subsets with no identified language, and takes the top $K - 1$. The lists are nested, i.e. the $K = 2$ list (Russian) is contained in the $K = 8$ list (Russian, Chinese, German, Japanese, Spanish, French and Italian), which is contained in the $K = 15$, $K = 30$ and $K = 50$ lists.

Scheme B lists. *Scheme B* prioritises diversity and generates language lists at $K \in \{8, 15, 30\}$. It first takes the largest subset of each script not yet covered, then the largest subset of each language family not yet covered, and then the remaining subsets in order of size. At $K = 8$, the scheme-B list spans seven scripts against four for scheme A. At $K = 30$, it spans 14 scripts and 12 families against the 10 and 10 from scheme A.

Schemes at $K \leq 2$. At $K = 2$, there is no language list to diversify, so the alternative builds change the second language. The scheme-B

build *ZH* replaces Russian with Chinese, and the scheme-C build *ES* replaces it with Spanish. At $K = 1$, there is no second language, so the alternative builds change the English corpus. The default is DCLM-Edu, the DCLM pipeline with an educational-quality filter. The scheme-B build *DCLMP* uses the same pipeline without that filter. The scheme-C build *FWEB* uses FineWeb restricted to the crawls up to 2022, which is DCLM's own Common Crawl window, so the two corpora do not differ in recency. All these alternative data builds are trained on the deep ladder from 90M to 1.7B at seed 1904 (Table 3).

Table 4 lists English and the 99 FineWeb-2 languages of the sweep, with the smallest language setting whose scheme-A and scheme-B lists contain each language and the number of benchmarks available in the harness for it.

Mixture composition. Every multilingual build blends English and FineWeb-2 50/50 at training time. The blend reads one English dataset and one FineWeb-2 dataset per build and language setting. Within the FineWeb-2 half, language i receives a share $q_i \propto p_i^{1/T}$, where p_i is its estimated token proportion in the corpus. At $T = 1$ each language receives its share of the corpus, and $T = 3$ flattens the distribution towards the smaller languages. The baseline build A uses $T = 1$, and the temperature intervention AT3 uses $T = 3$ on the same lists.

Validation set. Every model and every build share one fixed validation set. It covers English and the 99 FineWeb-2 languages with higher data availability. For each language, it takes the first 5M tokens of the first parquet file of that language, but at most 30% of the rows of the file, so that languages with a single file keep training data. The build records, for each language, the token count, the UTF-8 byte count and the number of leading rows assigned to validation. Every training build reads this record and skips exactly those rows, so training and validation data are disjoint by construction. Because one validation set serves the whole sweep, bits-per-byte is comparable across sizes, language settings and builds.

C Evaluation

We evaluate every model on bits-per-byte (BPB) of held-out text and on downstream benchmark tasks. Each task has one of three possible answer formats (MCQA-style letters, cloze, and generative continuation) and two scoring options (accuracy and benchmark bits-per-byte). All models are evaluated at the same aligned checkpoints: the ten tenths of training and two extra checkpoints at 85% and 95% for the noise estimate (Appendix A.3). Table 5 gives the notation.

C.1 Evaluation Configuration

C.1.1 Bits-per-byte (BPB)

BPB is the negative log-likelihood of held-out text in bits, normalized by its UTF-8 byte count. Lower BPB indicates better prediction of the held-out text. We include BPB as a reference signal, connecting our analysis to established loss-based scaling laws and providing a point of comparison for the scaling behaviour of downstream benchmarks.

For language ℓ, let $x_1, \ldots, x_{m_\ell}$ be the evaluated token sequence, c_t the context preceding token x_t within its scoring block, and b_ℓ the number of bytes in the evaluated text. We compute

$$\mathrm{BPB}_\ell = \frac{-\sum_{t=2}^{m_\ell} \log_2 p(x_t \mid c_t)}{b_\ell}. \qquad (1)$$

The evaluated text comes from the shared validation set of Appendix B. BPB connects the analysis to loss-based scaling laws and is a baseline for the downstream tasks.

C.2 Benchmarks

Benchmark selection. We considered 95 benchmarks, with 1,488 language tasks on the 50 trained languages. To try to increase the benchmark coverage of the low-resource languages, besides evaluating commonly used benchmarks such as ARC, HellaSwag or MMLU, we screened language-specific benchmarks like the Arabic Open LLM Leaderboard, FrenchBench, and IberoBench. To select which language-specific benchmarks to add before launching the evaluation of 12 checkpoints per model size, we evaluated the final checkpoints of the 600M, 1B, 1.7B, and 3B models, and discarded the 229 tasks that stayed at chance at both 1B and 1.7B (Figure 5, Appendix D.1). Table 6 details all the benchmarks considered, including information about the languages covered and the answer format of their tasks.

We evaluate 52 benchmarks for 50 languages, spanning reading comprehension, commonsense, factual knowledge, grammatical acceptability, and other capabilities (Table 6). We selected 938 tasks from the total 1,488 we considered, removing tasks for which the final checkpoints of the 600M, 1B,

Subset	Language	Script	Family	K_A	K_B	Fam.
dclm	English	Latn	Indo-European	1	1	25
rus_Cyrl	Russian	Cyrl	Indo-European	2	8	14
cmn_Hani	Mandarin Chinese	Hani	Sino-Tibetan	8	2	16
deu_Latn	German	Latn	Indo-European	8	8	13
jpn_Jpan	Japanese	Jpan	Japonic	8	8	10
spa_Latn	Spanish	Latn	Indo-European	8	30	18
fra_Latn	French	Latn	Indo-European	8	30	13
ita_Latn	Italian	Latn	Indo-European	8	30	12
por_Latn	Portuguese	Latn	Indo-European	15	30	11
pol_Latn	Polish	Latn	Indo-European	15	30	8
nld_Latn	Dutch	Latn	Indo-European	15	30	8
ind_Latn	Indonesian	Latn	Austronesian	15	30	10
vie_Latn	Vietnamese	Latn	Austro-Asiatic	15	30	13
fas_Arab	Persian	Arab	Indo-European	15	8	7
arb_Arab	Standard Arabic	Arab	Afro-Asiatic	15	30	14
tur_Latn	Turkish	Latn	Turkic	30	30	9
tha_Thai	Thai	Thai	Kra-Dai	30	8	4
ukr_Cyrl	Ukrainian	Cyrl	Indo-European	30	30	9
ell_Grek	Modern Greek (1453 onward)	Grek	Indo-European	30	8	9
kor_Hang	Korean	Hang	Koreanic	30	15	7
ces_Latn	Czech	Latn	Indo-European	30	30	6
swe_Latn	Swedish	Latn	Indo-European	30	30	9
hun_Latn	Hungarian	Latn	Uralic	30	30	9
ron_Latn	Romanian	Latn	Indo-European	30	30	6
nob_Latn	Norwegian Bokmål	Latn	Indo-European	30	–	3
dan_Latn	Danish	Latn	Indo-European	30	–	7
bul_Cyrl	Bulgarian	Cyrl	Indo-European	30	–	7
fin_Latn	Finnish	Latn	Uralic	30	–	7
hin_Deva	Hindi	Deva	Indo-European	30	15	13
ben_Beng	Bengali	Beng	Indo-European	30	15	9
slk_Latn	Slovak	Latn	Indo-European	50	–	7
heb_Hebr	Hebrew	Hebr	Afro-Asiatic	50	15	7
lit_Latn	Lithuanian	Latn	Indo-European	50	–	7
bos_Latn	Bosnian	Latn	Indo-European	50	–	1
slv_Latn	Slovenian	Latn	Indo-European	50	–	3
ekk_Latn	Standard Estonian	Latn	Uralic	50	–	7
cat_Latn	Catalan	Latn	Indo-European	50	–	8
tam_Taml	Tamil	Taml	Dravidian	50	15	9
hrv_Latn	Croatian	Latn	Indo-European	50	–	7
lvs_Latn	Standard Latvian	Latn	Indo-European	50	–	1
zsm_Latn	Standard Malay	Latn	Austronesian	50	–	6
azj_Latn	North Azerbaijani	Latn	Turkic	50	–	5
srp_Cyrl	Serbian	Cyrl	Indo-European	50	–	8
kat_Geor	Georgian	Geor	Kartvelian	50	15	6
npi_Deva	Nepali (individual language)	Deva	Indo-European	50	–	8
mar_Deva	Marathi	Deva	Indo-European	50	–	7
mal_Mlym	Malayalam	Mlym	Dravidian	50	15	7
kaz_Cyrl	Kazakh	Cyrl	Turkic	50	–	6
urd_Arab	Urdu	Arab	Indo-European	50	–	7
als_Latn	Tosk Albanian	Latn	Indo-European	50	–	5
mkd_Cyrl	Macedonian	Cyrl	Indo-European	val	–	4
tel_Telu	Telugu	Telu	Dravidian	val	–	7
kan_Knda	Kannada	Knda	Dravidian	val	–	4
mya_Mymr	Burmese	Mymr	Sino-Tibetan	val	–	2
guj_Gujr	Gujarati	Gujr	Indo-European	val	–	5
bel_Cyrl	Belarusian	Cyrl	Indo-European	val	–	3
isl_Latn	Icelandic	Latn	Indo-European	val	–	3
khm_Khmr	Khmer	Khmr	Austro-Asiatic	val	–	1
khk_Cyrl	Halh Mongolian	Cyrl	Mongolic	val	–	1
fil_Latn	Filipino	Latn	Austronesian	val	–	3
ary_Arab	Moroccan Arabic	Arab	Afro-Asiatic	val	–	14
afr_Latn	Afrikaans	Latn	Indo-European	val	–	1
hye_Armn	Armenian	Armn	Indo-European	val	–	6
sin_Sinh	Sinhala	Sinh	Indo-European	val	–	3
glg_Latn	Galician	Latn	Indo-European	val	–	6
uzn_Cyrl	Northern Uzbek	Cyrl	Turkic	val	–	3
pan_Guru	Panjabi	Guru	Indo-European	val	–	2
ory_Orya	Odia	Orya	Indo-European	val	–	1
uzn_Latn	Northern Uzbek	Latn	Turkic	val	–	3
kir_Cyrl	Kirghiz	Cyrl	Turkic	val	–	4
eus_Latn	Basque	Latn	Language isolate	val	–	10
lat_Latn	Latin	Latn	Indo-European	val	–	1
tgk_Cyrl	Tajik	Cyrl	Indo-European	val	–	1
swh_Latn	Swahili (individual language)	Latn	Niger-Congo	val	–	6
arz_Arab	Egyptian Arabic	Arab	Afro-Asiatic	val	–	14
nno_Latn	Norwegian Nynorsk	Latn	Indo-European	val	–	1
cym_Latn	Welsh	Latn	Indo-European	val	–	1
amh_Ethi	Amharic	Ethi	Afro-Asiatic	val	–	4
pbt_Arab	Southern Pashto	Arab	Indo-European	val	–	1
ckb_Arab	Central Kurdish	Arab	Indo-European	val	–	2
ars_Arab	Najdi Arabic	Arab	Afro-Asiatic	val	–	14
gle_Latn	Irish	Latn	Indo-European	val	–	1
lao_Laoo	Lao	Laoo	Kra-Dai	val	–	1
srp_Latn	Serbian	Latn	Indo-European	val	–	8
som_Latn	Somali	Latn	Afro-Asiatic	val	–	2
bod_Tibt	Tibetan	Tibt	Sino-Tibetan	val	–	1
mlt_Latn	Maltese	Latn	Afro-Asiatic	val	–	1
asm_Beng	Assamese	Beng	Indo-European	val	–	2
snd_Arab	Sindhi	Arab	Indo-European	val	–	2
kmr_Latn	Northern Kurdish	Latn	Indo-European	val	–	1
uig_Arab	Uighur	Arab	Turkic	val	–	2
plt_Latn	Plateau Malagasy	Latn	Austronesian	val	–	2
kin_Latn	Kinyarwanda	Latn	Niger-Congo	val	–	2
jav_Latn	Javanese	Latn	Austronesian	val	–	2
xho_Latn	Xhosa	Latn	Niger-Congo	val	–	1
hat_Latn	Haitian	Latn	Creole	val	–	2
fao_Latn	Faroese	Latn	Indo-European	val	–	2
zul_Latn	Zulu	Latn	Niger-Congo	val	–	2
ibo_Latn	Igbo	Latn	Niger-Congo	val	–	2
sot_Latn	Southern Sotho	Latn	Niger-Congo	val	–	1

Table 4: **Training languages of the model ladder.** List of languages with their script and family annotated, as well as the name of the corresponding FineWeb2 subset. K_A gives the smallest language setting whose scheme-A (resource-ranked) list contains the subset. K_B gives the smallest language setting of a scheme-B build that contains it. The lists are nested, so a language is also trained at every larger setting of its scheme. The 50 languages marked “val" are in no training mixture, they are only included in the 100-language validation set. Fam. counts the benchmark families of the evaluation suite (reformulated variants excluded) that the harness offers for the language.

Symbol	Meaning	Example
b	Benchmark	HellaSwag
ℓ	Evaluation language	Spanish
b_ℓ	Language task	HellaSwag-es
$b_{\ell i}$	Item	One question of HellaSwag-es
Y_{b_ℓ}	Task score	accuracy, bBPB or BPB
B	Evaluation suite	All benchmarks

Table 5: Notation for the evaluation.

1.7B and 3B models did not achieve above-chance performance.

Task selection. A model is evaluated on the tasks whose language is in its training mixture. The number of tasks evaluated for each model grows with the language count K: 113 at $K = 1$, 134 at $K = 2$, 326 at $K = 8$, 485 at $K = 15$, 720 at $K = 30$, and 938 at $K = 50$. These counts cover every group of Table 6: the multilingual families, the language-specific benchmarks, and the reformulated variants of the letter-format families (§4.2). Some runs are also evaluated on every task in every language, regardless of their training mixture. These are the deep seed-1904 runs of data build A at every K, and all language schemes at $K = 2$ and $K = 8$.

Language coverage. Every language of the $K = 50$ list is covered by at least one benchmark, but coverage is scarce towards the low-resource end of the list (Table 4). English is covered by 31 benchmarks, 17 multilingual and 14 English-only. The other 49 languages have a median of eight benchmarks. The most covered are Spanish (18), Mandarin Chinese (17), Russian and Standard Arabic (14 each), and Hindi, Italian and Vietnamese (13 each). The least covered are mostly languages that enter only at $K = 50$: Bosnian has only Global PIQA, Standard Latvian has Belebele and MultiBLiMP, and Slovenian has three families (Belebele, Global PIQA, and MultiBLiMP). To try to increase language coverage, besides evaluating commonly used benchmarks such as ARC, HellaSwag or MMLU, we screened language-specific benchmarks like the Arabic Open LLM Leaderboard, FrenchBench, IberoBench. However, most of the low-resource benchmarks didn't pass the above chance threshold at 1B or 1.7B and weren't included in the analysis.

C.3 Answer Formats and Scoring

Answer formats. A task can follow one of three formats. A *letter* task lists the options in the prompt with labels (A, B, . . .) and scores the label. A *cloze* task scores each answer text as a continuation of the prompt. A *completion* task has no shared prompt (minimal pairs, sentence pairs, Winograd schemas) or a single target continuation (LAMBADA). For our analyses, we reformat the letter-format tasks either programmatically or using an LLM to convert them into cloze tasks (§4).

Harness. We use the LM Evaluation Harness v0.4x (Gao et al., 2024) and use the original prompt and few-shot examples. We score accuracy (`acc`), and length-normalised accuracy (`acc_norm`) for the benchmarks with answers in cloze format. By default, we evaluate each model only on the languages included in its training mixture.

Accuracy Tasks whose options are answer strings of different lengths use length-normalised accuracy (`acc_norm`): the cloze-format originals (e.g. ARC, HellaSwag, OpenBookQA, MathQA and Global PIQA), and the reformulated versions for letter-format benchmarks (e.g. INCLUDE, Belebele, Global-MMLU). The other tasks use accuracy (`acc`), the harness default. On letter-format tasks the two are identical, since every option is a single label of the same length.

Benchmark bits-per-byte (bBPB) For every benchmark with a gold answer, we also score the gold answer by its log-likelihood, normalized by its length in UTF-8 bytes. Lower bBPB is better, and it has no chance level. We propose it as measurement where benchmark accuracy is not above chance. For an item x with gold answer a^*, the Harness gives the log-likelihood $\log p(a^* \mid x)$ of the gold continuation in nats. Over the n items of a task, with $|a^*|$ being the length of the gold continuation in UTF-8 bytes,

$$\mathrm{bBPB}_{b_\ell} = \frac{1}{n} \sum_{x} \frac{-\log p(a^* \mid x)}{\ln 2 \cdot |a^*|}.$$

Benchmark	Langs.	Trained	Task	Formats	Mid-Resource Langs.
Analysed (40 benchmarks, 695 tasks)					
ACP-Bench (Kokel et al., 2025)	1	2	Reasoning	2-4-option letter/cloze, RF	–
ARC (MT) (LumiOpen, 2024)	11	11	Knowledge	4-option cloze	–
ARC (Okapi) (Lai et al., 2023)	32	27	Knowledge	4-option cloze	ml mr ne sr hr ta ca sk bn
BBH (Suzgun et al., 2023)	1	2	Reasoning	2–10-option letter/cloze, RF	–
Belebele (Bandarkar et al., 2024)	94	49	Reading	4-option letter, RF, LLM-RF	sq ur kk ml mr ne ka sr az ms lv hr ta ca et sl lt he sk bn
BLEnD (Myung et al., 2024)	4	4	Cultural knowledge	4-option letter	–
BoolQ (Clark et al., 2019)	1	1	Reading	2-option cloze	–
CommonsenseQA (Talmor et al., 2019)	1	1	Commonsense	5-option letter, RF	–
CulturalBench (Chiu et al., 2025)	8	16	Cultural knowledge	4-option letter, RF	–
Global PIQA (Chang et al., 2026)	77	48	Commonsense	2–4-option cloze	sq ur kk ml mr ne ka sr az ms hr ta ca et sl bs lt he sk bn
Global-MMLU (Singh et al., 2025)	37	29	Knowledge	4-option letter, RF	ne sr ms lt he bn
HellaSwag (Okapi) (Lai et al., 2023)	31	26	Commonsense	4-option cloze	ml mr ne sr hr ta ca sk bn
INCLUDE (Romanou et al., 2025)	43	36	Knowledge	4-option letter, RF, LLM-RF	sq ur kk ml ne ka sr az ms hr ta et lt he bn
INCLUDE v2 (Romanou et al., 2025)	42	42	Knowledge	4-option cloze	sq ur kk ml mr ne ka sr az ms hr ta et lt he sk bn
KoBEST (Jang et al., 2022)	1	1	Mixed suite	2–4-option cloze	–
LAMBADA (Gao et al., 2024)	5	5	Language modelling	completion	–
MathQA (Amini et al., 2019)	1	1	Reasoning	5-option cloze	–
MMLU (Hendrycks et al., 2021)	1	1	Knowledge	4-option letter, RF	–
MMLU (Okapi) (Lai et al., 2023)	27	27	Knowledge	4-option letter, RF	ml mr ne sr hr ta ca sk bn
MMMLU (OpenAI, 2024)	12	12	Knowledge	4-option letter, RF	bn
MultiBLiMP (Jumelet et al., 2026)	60	37	Grammar	completion	sq ur kk mr ka lv hr ta ca et sl lt he sk bn
NorEval (Mikhailov et al., 2025)	1	1	Mixed suite	2–5-option letter/cloze, completion	–
OpenBookQA (Mihaylov et al., 2018)	1	1	Knowledge	4-option cloze	–
PAWS-X (Yang et al., 2019)	10	8	Inference	completion	ca
PIQA (Bisk et al., 2020)	1	1	Commonsense	2-option cloze	–
TokSuite (Altıntaş et al., 2026)	5	5	Tokenization	4-option cloze	–
ToxiGen (Hartvigsen et al., 2022)	1	1	Toxicity	2-option cloze	–
TruthfulQA (MC2) (Lin et al., 2022a; Lai et al., 2023)	3	3	Truthfulness	var.-option cloze	–
TruthfulQA-Multi (MC1) (Calvo Figueras et al., 2025)	3	2	Truthfulness	var.-option cloze	–
WinoGrande (Sakaguchi et al., 2020)	1	1	Commonsense	completion	–
XCOPA (Ponti et al., 2020)	11	8	Commonsense	2-option cloze	ta et
XNLI (Conneau et al., 2018)	18	15	Inference	completion	ur ca
XStoryCloze (Lin et al., 2022b)	13	8	Commonsense	2-option cloze	ca
XWinograd (Tikhonov and Ryabinin, 2021)	6	6	Commonsense	completion	–
Screened, not analysed (26 benchmarks, 485 tasks)					
ACVA (OALL) (Huang et al., 2024)	1	1	Cultural knowledge	2-option cloze	–
AlGhafa (OALL) (Almazrouei et al., 2023)	1	1	Mixed suite	2–5-option cloze	–
AlGhafa translated (OALL) (El Filali et al., 2024)	1	1	Mixed suite	2–4-option cloze	–
Arabic EXAMS (OALL) (Hardalov et al., 2020)	1	1	Knowledge	4-option letter, RF	–
Arabic MMLU (OALL) (El Filali et al., 2024)	1	1	Knowledge	4-option letter, RF	–
ArabicMMLU (Koto et al., 2024)	1	1	Knowledge	var.-option letter, RF	–
Bangla (Nahin et al., 2025)	1	1	Mixed suite	2–5-option letter	bn
BLiMP-NL (Suijkerbuijk et al., 2025)	1	1	Grammar	completion	–
C-Eval (Huang et al., 2023)	1	1	Knowledge	4-option letter	–
CareQA (Arias-Duart et al., 2025)	1	1	Knowledge	4-option letter, RF	–
COPAL-ID (Wibowo et al., 2024)	1	1	Commonsense	2-option cloze	–
EVALITA-LLM (Magnini et al., 2025)	1	1	Mixed suite	2–5-option letter/cloze	–
FrenchBench (Faysse et al., 2025)	1	1	Mixed suite	2–4-option letter/cloze, RF	–
HAE-RAE Bench (Son et al., 2024)	1	1	Knowledge	5-option letter	–
HEAD-QA (Vilares and Gómez-Rodríguez, 2019)	1	1	Knowledge	4-option cloze	–
IberoBench (Baucells et al., 2025)	5	3	Mixed suite	2–4-option cloze	ca
KMMLU (Son et al., 2025)	1	1	Knowledge	4-option letter	–
MELA (Zhang et al., 2024)	10	9	Grammar	2-option letter	–
TruthfulQA (MC1) (Lin et al., 2022a)	1	1	Truthfulness	var.-option cloze	–
TruthfulQA (Okapi, MC1) (Lai et al., 2023)	27	26	Truthfulness	var.-option cloze	ml mr ne sr hr ta ca sk bn
TruthfulQA (Okapi, MC2) (Lai et al., 2023)	24	24	Truthfulness	var.-option cloze	ml mr ne sr hr ta ca sk bn
TruthfulQA-Multi (MC2) (Calvo Figueras et al., 2025)	3	3	Truthfulness	var.-option cloze	ca
TurBLiMP (Başar et al., 2025)	1	1	Grammar	completion	–
TurkishMMLU (Yüksel et al., 2024)	1	1	Knowledge	5-option letter	–
XQuAD (Artetxe et al., 2020)	1	1	Reading	completion	–
ZhoBLiMP (Liu et al., 2026)	1	1	Grammar	completion	–

Table 6: **Every benchmark considered for the analysis.** Langs is the number of languages that the harness registers for the benchmark. Trained is the number of languages included in our model ladder (English and the FineWeb-2 languages of every trained build). Formats lists the scoring formats of the original tasks with n answer options, a range when n differs across tasks, and var. when it only varies across items; "letter" scores the labels (A, B, . . .) of options listed in the prompt, "cloze" scores each answer text as a continuation of the prompt, and "completion" has no shared prompt (minimal pairs, sentence pairs, Winograd schemas). Mid-Resource Language Examples lists the benchmark's languages among ranks 30 to 50 of our 50-language list, which orders FineWeb-2 languages by resources, from the least resourced up (ISO 639-1 codes).

D Measurability Analyses

This appendix details the analyses of §4: the above-chance threshold (§D.1), the reformulation of letter-format tasks (§D.2), the training tokens of a language needed to pass the threshold (§D.3) and scaling predictability (§D.4).

D.1 The Above-Chance Threshold

Definition. For each language task b_ℓ and model size N, we test whether the observed accuracy is distinguishable from random guessing. For a task with m_b answer options, chance accuracy is $1/m_b$. Where the option count varies across items, chance is the mean of $1/m$ over the items (0.2253 for TruthfulQA MC1). For every model of size N, we take its final-checkpoint accuracy $\hat{p}$ over the n scored items of the task. We recover the number of correct answers as $\mathrm{round}(\hat{p}\,n)$ and compute the one-sided 95% Wilson lower confidence bound on the proportion. The model is above chance when

$$\mathrm{LCB}_{95}(\hat{p}) > \frac{1}{m_b}.$$

We then aggregate these decisions over the models of the same size, i.e., every ladder and every data build at seed 1904. A (b_ℓ, N) pair passes the threshold if at least 50% of these models are above chance. Only the models whose mixture trains the task's language are considered. The resulting mask is the gate that every analysis reads. A scaling fit uses only the sizes where the task passes. A ranking against the reference also requires the task to pass at the reference size. Measurements without a meaningful chance baseline, such as BPB and bBPB, are not filtered by this procedure.

Per-benchmark required margin. The Wilson bound accounts for the number of evaluation examples, so the amount by which accuracy must exceed chance varies across tasks. Smaller evaluation sets require a larger margin above chance, while larger sets can support smaller but more reliable departures from chance. For example, the median required margins are 9.00 percentage points for nonparallel Global PIQA (100 items per language), 2.44 for Belebele (900), 0.74 for HellaSwag (about 9,000), and 0.60 for Global-MMLU (about 14,000).

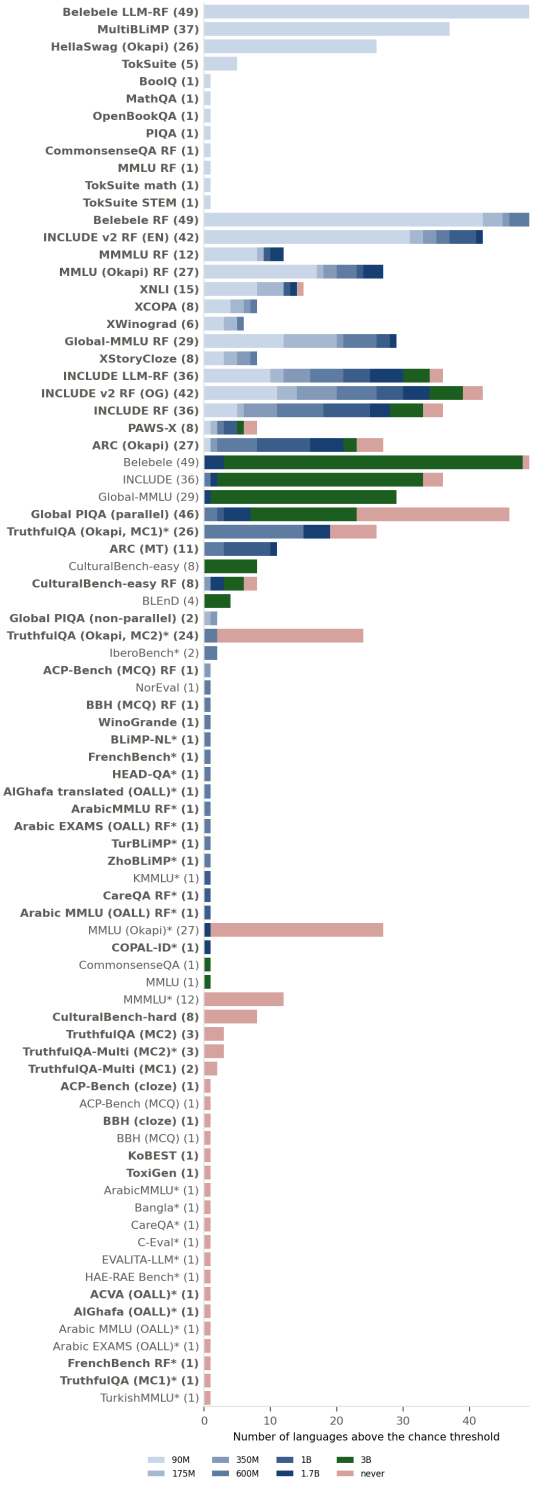


Figure 5: **Most letter-format benchmarks pass the above chance threshold only at 3B, while cloze benchmarks pass from 90M.** For every benchmark we considered, the bar corresponds to the number of trained languages in which it is above chance, coloured by the smallest model size from which it passes. Benchmarks in bold are scored in cloze format. A * marks a benchmark that we screened but did not analyse.

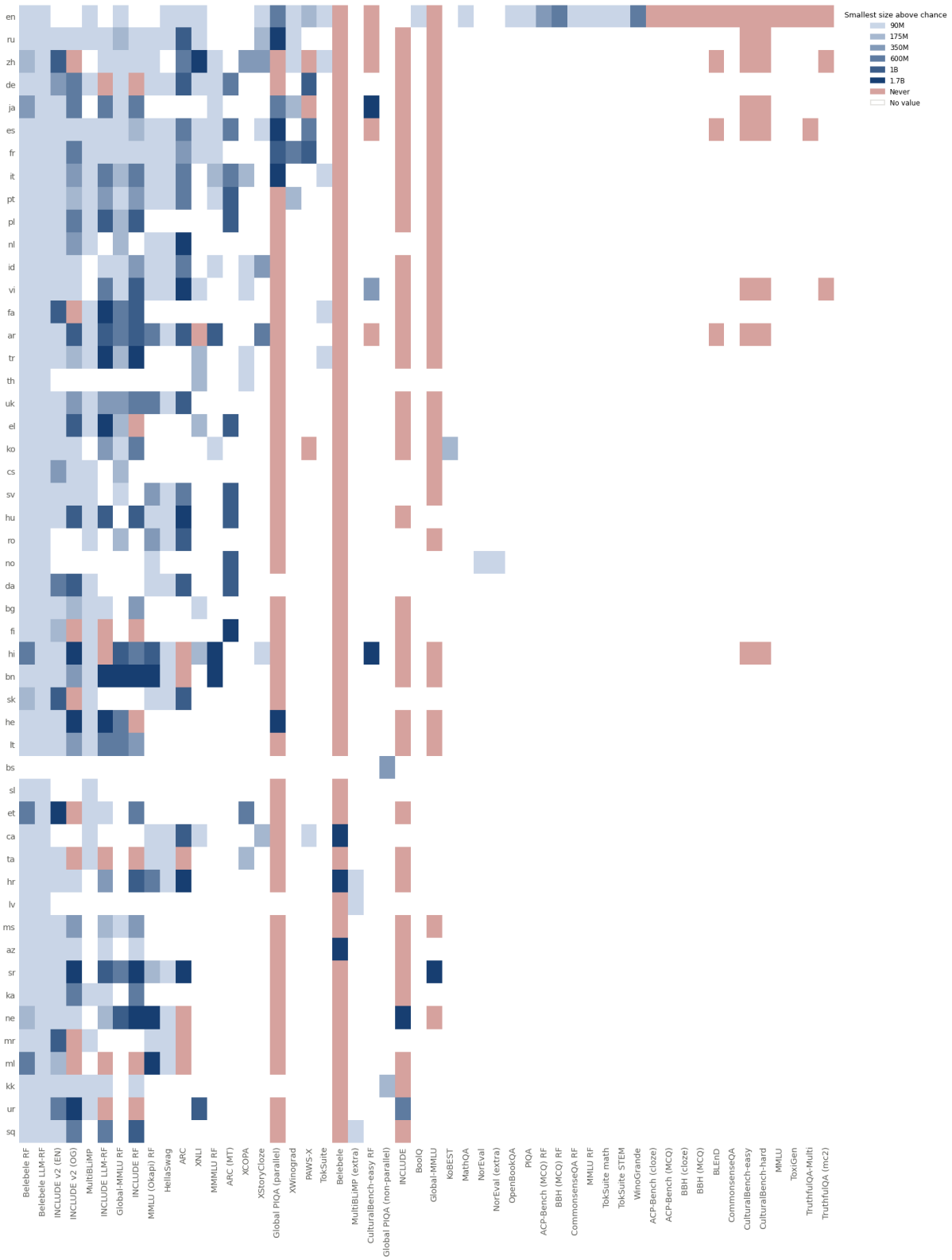


Figure 6: **Language tasks that pass the above-chance threshold, per benchmark and model size.** The screened benchmarks that we do not analyse are left out.

Majority-class chance level. When the gold answers of a task are unbalanced, always picking the most frequent answer position can beat $1/m_b$. We therefore rerun the threshold with the larger of the uniform level and this majority baseline as the chance level. Applied to the labelled tasks, it lowers the share of tasks above chance by 1.5 to 2.8 percentage points, depending on the size. The tasks it removes are the Belebele, Global-MMLU and INCLUDE tasks that barely pass the uniform threshold, CulturalBench-easy, PAWS-X and BLEnD.

Above-chance results per benchmark. Figure 5 gives, for every benchmark we considered, the number of trained languages in which it passes the threshold and the smallest size from which it does. Figure 6 gives the full grid per benchmark and language.

D.2 Task Reformatting and Reformulation

We test whether the letter-format benchmarks pass the above-chance threshold as published and after two reformulations of the same items.

The programmatic reformulation (RF) drops the answer letters and scores each option as a continuation of the question. The model-based reformulation (LLM-RF) has `gemini-3.8-flash` (Google DeepMind, 2026) rewrite every item into a cloze statement whose options complete it, without revealing the correct option (Belebele and INCLUDE only).

Each reformulated task goes through the same threshold as its original, on the same seed-1904 runs from 90M to 1.7B. A McNemar test on the paired verdicts of an original and its reformulation, one pair per task, tests the difference at every size.

Dropping the letters moves whole benchmarks above chance. At 1.7B, RF passes in 86 of 105 Belebele, 35 of 37 Global-MMLU and 31 of 43 INCLUDE languages, against 5, 1 and 3 for the originals (McNemar $p \leq 5.8 \times 10^{-8}$). The difference is significant at every size for Belebele and Global-MMLU and from 175M for INCLUDE.

The LLM rewrite further improves results. LLM-RF Belebele passes in every trained language at every size, and LLM-RF INCLUDE in 32 of 43 languages at 1.7B. The rewrite introduces no length cue: the correct option is the single longest in 22.2% of the original items and 21.9% of the rewritten ones, against 25% by chance. Figure 7 gives the results per benchmark and size.

D.3 Above chance against the training tokens of the language

We measure the share of evaluations above chance, where an evaluation is one (task, size, K) at one of the ten evaluated tenths of a run. We use the seed-1904 runs of every data build and ladder, from 90M to 1.7B, and the benchmark tasks with a chance level in trained languages.

An evaluation is above chance when it passes the threshold at that checkpoint, and it is placed at the number of tokens of the task's language that the checkpoint has seen: the language's share of the mixture times the token budget times the share of the run. Per benchmark, answer format and model size, we fit a logistic curve of the evaluations' verdict on the log of these tokens, using the non-English evaluations, and read the token counts at which it reaches 50% and 90% of the evaluations (Table 7).

For non-English tasks, the share above chance rises from 100M tokens of the language. The share of evaluations above chance stays near 40% up to 100M tokens of the language and rises with every decade after: 43% at 10^5 (82 evaluations), 42% at 1M and 10M, 52% at 100M, 62% at 1B and 64% at 10B (4,945 evaluations) (Figure 8.)

Exposure to training data is necessary but not sufficient. At 100M–1B tokens, 43% of the 90M evaluations are above chance against 62% of the 1.7B evaluations. Pooled over every language, the curve instead peaks at 100M–1B tokens (52%) and falls to 41% at 10B–100B, because the top bins are mostly English (80% of the 25,217 evaluations). In the thresholds of Table 7, ten of the 30 benchmark formats never reach 50% of their evaluations above chance, among them the letter-format Belebele, Global-MMLU and INCLUDE, while their reformulations do. Where a benchmark reaches the threshold, a 1.7B model needs about ten times fewer tokens of the language than a 90M one: ARC (Okapi), for example, reaches 50% at 3.3B tokens at 600M and at 320M tokens at 1.7B. Figure 8 gives the curves of every benchmark.

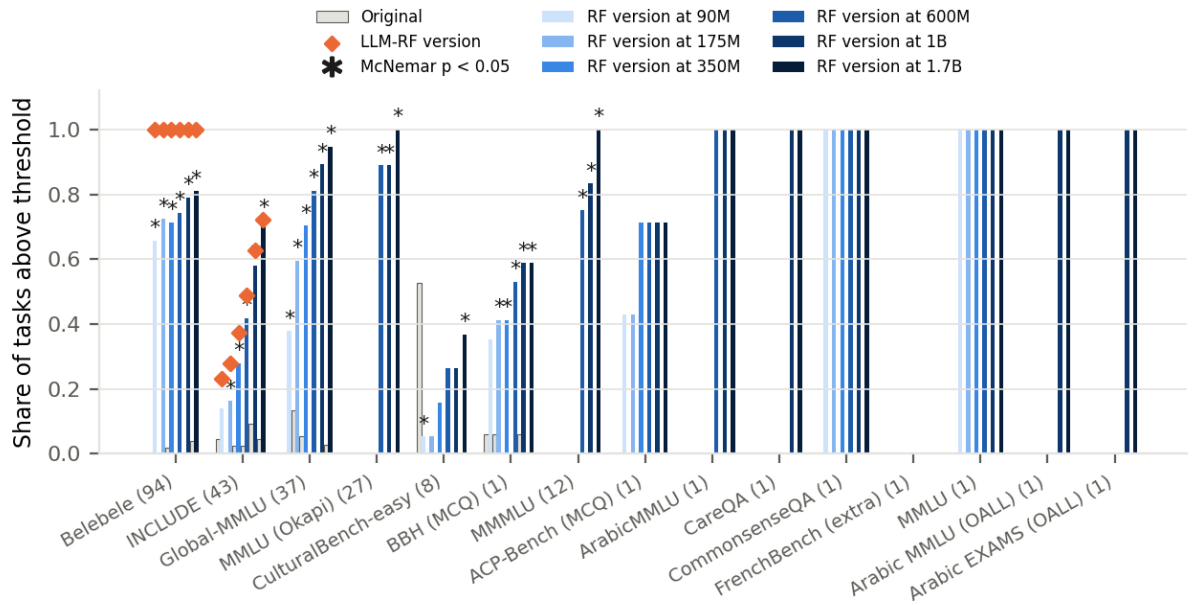


Figure 7: **Dropping the letters moves whole families across the gate.** For each letter-format benchmark (its number of tasks in parentheses), the share of tasks above the chance threshold for the original items (grey bars), for the RF version at each model size (blue bars, a darker blue is a larger size) and for the LLM-RF version (orange diamonds). A star marks a size at which the McNemar test between the original and its twin gives $p < 0.05$.

			Training tokens of the language for 50% / 90% of the evaluations above chance							
Benchmark	Format	Lang.	90M	175M	350M	600M	1B	1.7B	Margin	Items
ARC (MT)	Original	11	>4.6B	>8.8B	>17B	2.3B / 15B	500M / 1.8B	190M / 900M	0.021	1,172
ARC (Okapi)	Original	26	>4.6B	>8.8B	>17B	3.3B / >30B	720M / 3.9B	320M / 1.4B	0.021	1,169
Belebele	Original	48	>4.6B	>8.8B	>17B	>30B	>47B	>84B	0.024	900
	RF	48	≤440K / >4.6B	≤860K / 5.6B	≤1.7M / 550M	≤3M / 210M	≤4.6M / 51M	≤8.2M / 9M	0.024	900
	LLM-RF	48	≤440K	≤860K	≤1.7M	≤3M	≤4.6M	≤8.2M	0.024	898
BLEnD	Original	3	>4.6B	>8.8B	>17B	>30B	>47B	>84B	0.032	500
CulturalBench-easy	Original	7	≤3.1M / >4.6B	>8.8B	>17B	>30B	>47B	>84B	0.138	32
	RF	7	>4.6B	>8.8B	>17B	>30B	>47B	>84B	0.142	32
CulturalBench-hard	Original	7	>4.6B	>8.8B	>17B	>30B	>47B	>84B	0.077	130
Global PIQA (parallel)	Original	45	>4.6B	>8.8B	>17B	>30B	>47B	17B / >84B	0.070	103
Global-MMLU	Original	28	>4.6B	>8.8B	>17B	>30B	>47B	>84B	0.006	14,042
	RF	28	51M / 3.3B	55M / 600M	83M / 600M	88M / 550M	64M / 400M	≤39M / 210M	0.006	14,042
HellaSwag (Okapi)	Original	25	≤440K	≤860K	≤1.7M	≤3M	≤4.6M	≤8.2M	0.007	9,272
INCLUDE	Original	36	>4.6B	>8.8B	>17B	>30B	>47B	>84B	0.032	548
	RF	36	1.3B / >4.6B	1.1B / >8.8B	730M / 12B	400M / 8.1B	310M / 7.7B	150M / 4.9B	0.032	548
	LLM-RF	36	750M / >4.6B	520M / >8.8B	430M / >17B	250M / 13B	180M / 10B	79M / 4.9B	0.032	546
INCLUDE v2 RF (EN)	RF	42	≤440K / >4.6B	≤860K / >8.8B	≤1.7M / >17B	≤3M / >30B	≤4.6M / >47B	≤8.2M / >84B	0.024	886
INCLUDE v2 RF (OG)	RF	42	880M / >4.6B	880M / >8.8B	510M / >17B	310M / >30B	200M / >47B	92M / >84B	0.024	886
MMLU (Okapi)	RF	27	980K / >4.6B	9.4M / 680M	15M / 830M	19M / 450M	23M / 360M	16M / 230M	0.006	13,101
MMMLU	RF	12	30M / 190M	47M / 240M	84M / 420M	120M / 650M	140M / 680M	93M / 530M	0.006	14,042
MultiBLiMP	Original	33	≤570K	≤1.1M	≤2.2M	≤3.8M	≤6M	≤11M	0.017	2,298
PAWS-X	Original	7	>4.6B	>8.8B	>17B	>30B	>47B	11B / >84B	0.018	2,000
TokSuite	Original	4	≤6.4M / 13M	≤12M	≤24M	≤42M	≤66M	≤120M	0.125	40
TruthfulQA (MC2)	Original	2	>4.6B	>8.8B	>17B	>30B	>47B	>84B	0.030	786
XCOPA	Original	8	>4.6B	>8.8B	≤2.6M / >17B	13M / 260M	≤7.2M / 44M	≤13M / 14M	0.038	500
XNLI	Original	14	71M / >4.6B	≤4.4M / >8.8B	≤8.8M / >17B	≤15M / >30B	≤24M / >47B	≤43M / >84B	0.016	2,490
XStoryCloze	Original	7	>4.6B	500M / >8.8B	98M / 17B	≤20M / 700M	≤32M / 190M	≤56M	0.022	1,511
XWinograd	Original	5	290M / 1.6B	280M / 1.9B	270M / 2.4B	110M / 1B	≤140M / 590M	≤240M	0.049	315

Table 7: **Training tokens of the language a model of each size needs before a benchmark is above chance.** For each non-English multilingual benchmark and answer format, the training tokens of the language at which 50% / 90% of its (language, K, checkpoint) evaluations are above chance. “Lang." is the number of non-English trained languages, “Margin" the median accuracy above chance that a task needs for its Wilson bound to clear chance, and “Items" its median item count.

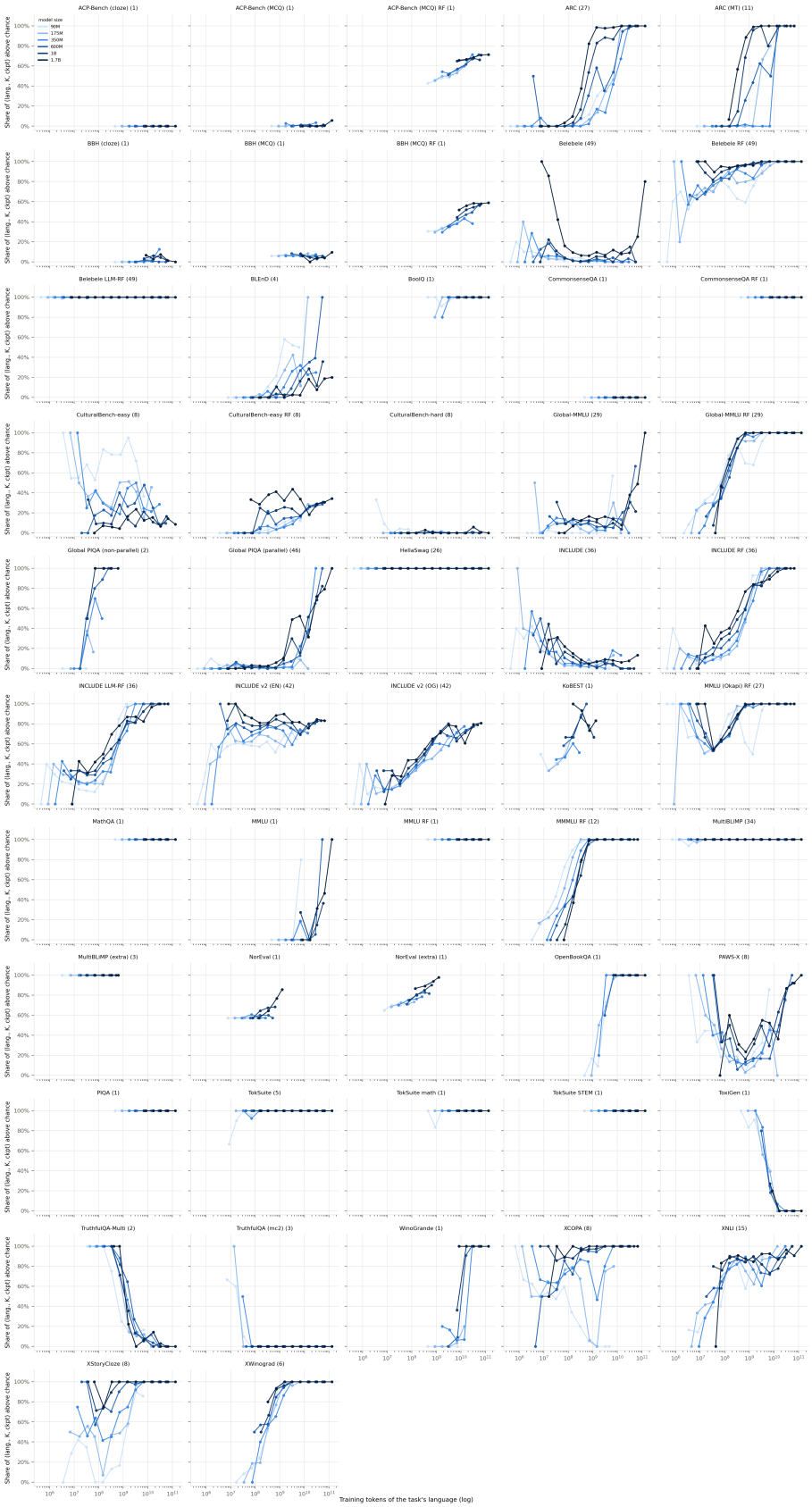


Figure 8: **For non-English tasks, the share of evaluation results above chance rises with every decade of the language's tokens from 10M on.** One panel per benchmark (its number of tasks in parentheses). Each point is the share of (language, K, checkpoint) cells above chance, binned by the training tokens of the task's language that the checkpoint had seen (log scale). There is one line per model size, a darker blue for a larger size.

D.4 Scaling Predictability

Scaling fit methodology. To assess scaling behaviour across model sizes, we consider sets of models that differ only in parameter count, holding all other training configuration choices fixed. For each set, we fit final-checkpoint task performance against $\log_{10} N$, where N denotes the number of model parameters. To assess scaling throughout training, we fit task performance against the logarithm of the number of training tokens consumed at each checkpoint, separately for each model and requiring at least five checkpoints. For each benchmark–language task, we restrict the analysis to training configurations that include the corresponding language in their training mixture and satisfy the above-chance filtering criterion.

We quantify how good each fit is using the coefficient of determination (R^2) and report Spearman's rank correlation (ρ) to characterize the monotonicity and direction of performance changes. For BPB and bBPB, where lower scores indicate better performance, we reverse the sign of ρ so that positive values consistently indicate improvement with increasing model size or training tokens.

The accuracy of the cloze versions of letter-format benchmarks increases steadily with both model size and along training. While many of the original letter-format versions of MCQ benchmarks stay at chance, their cloze versions scale predictably together with other originally cloze benchmarks. (Figures 9 and 10).

The commonly used English benchmarks scale predictably . To verify how the common English benchmarks behave without a multilingual mixture, we repeat the scaling predictability analysis on the monolingual English models ($K = 1$). These were trained on three data mixtures: the default DCLM-Edu English, DCLM without the educational filter, and FineWeb. In all three, 17 of the 39 English accuracy tasks are predictable along both size and training. These include the benchmarks commonly used in model development: ARC (easy and challenge), HellaSwag and OpenBookQA, together with the cloze forms of MMLU, Global-MMLU, Belebele and CommonsenseQA, whose original letter format never passes the filter. Global PIQA is the only popular benchmark above chance that does not scale predictably (Figure 11). These single-cell fits are not comparable in level with the medians over K of Figure 2.

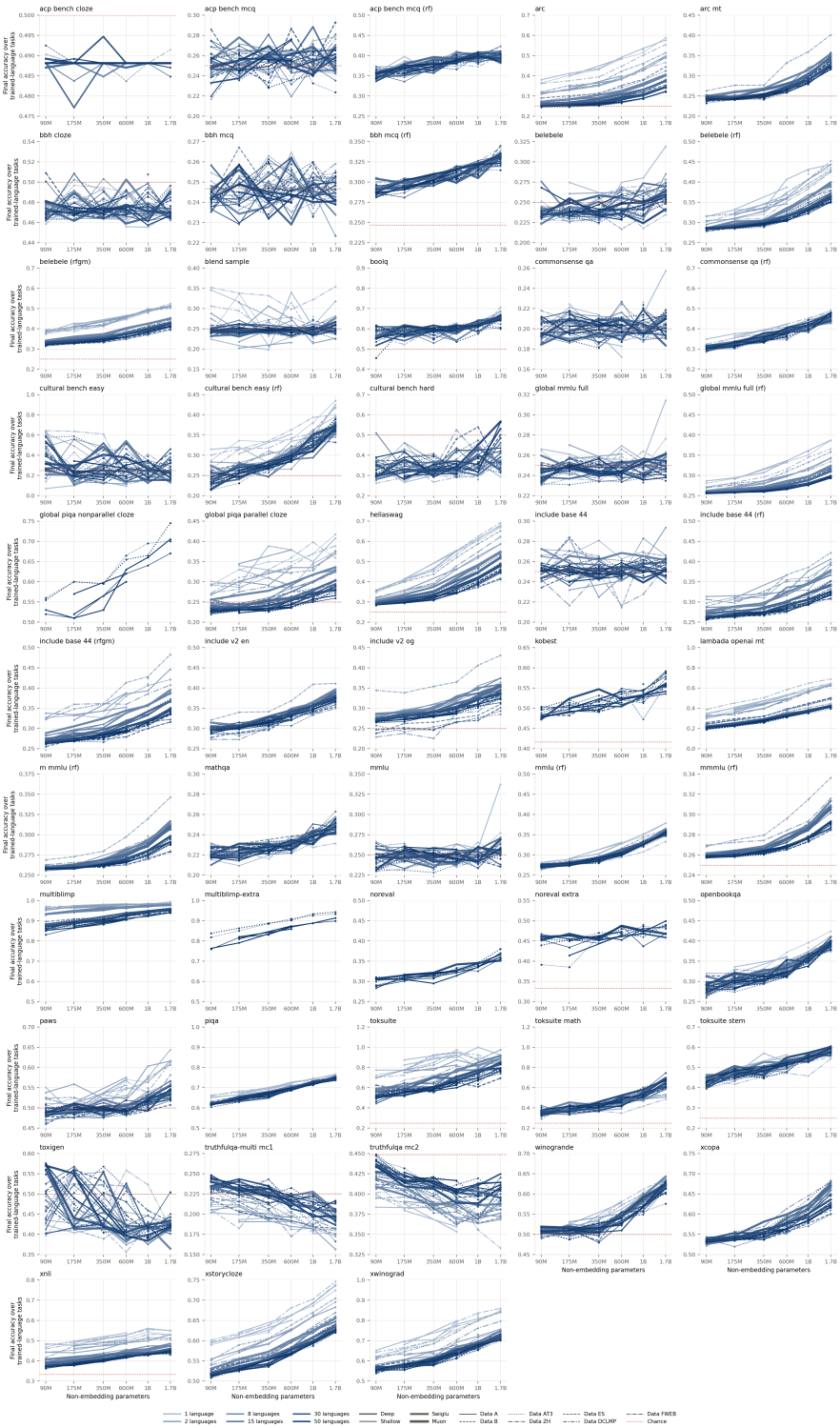


Figure 9: **The reformulated versions rise with model size while the letter-format original benchmarks stay at chance up to 1.7B.** Final-checkpoint accuracy against non-embedding parameters, averaged over the tasks in the languages each configuration trains. There is one line per (K, ladder, data build, seed) tuple, the colour indicates K, width the ladder and style the data build. The dotted red line represents chance. These curves are not filtered by the above-chance threshold.

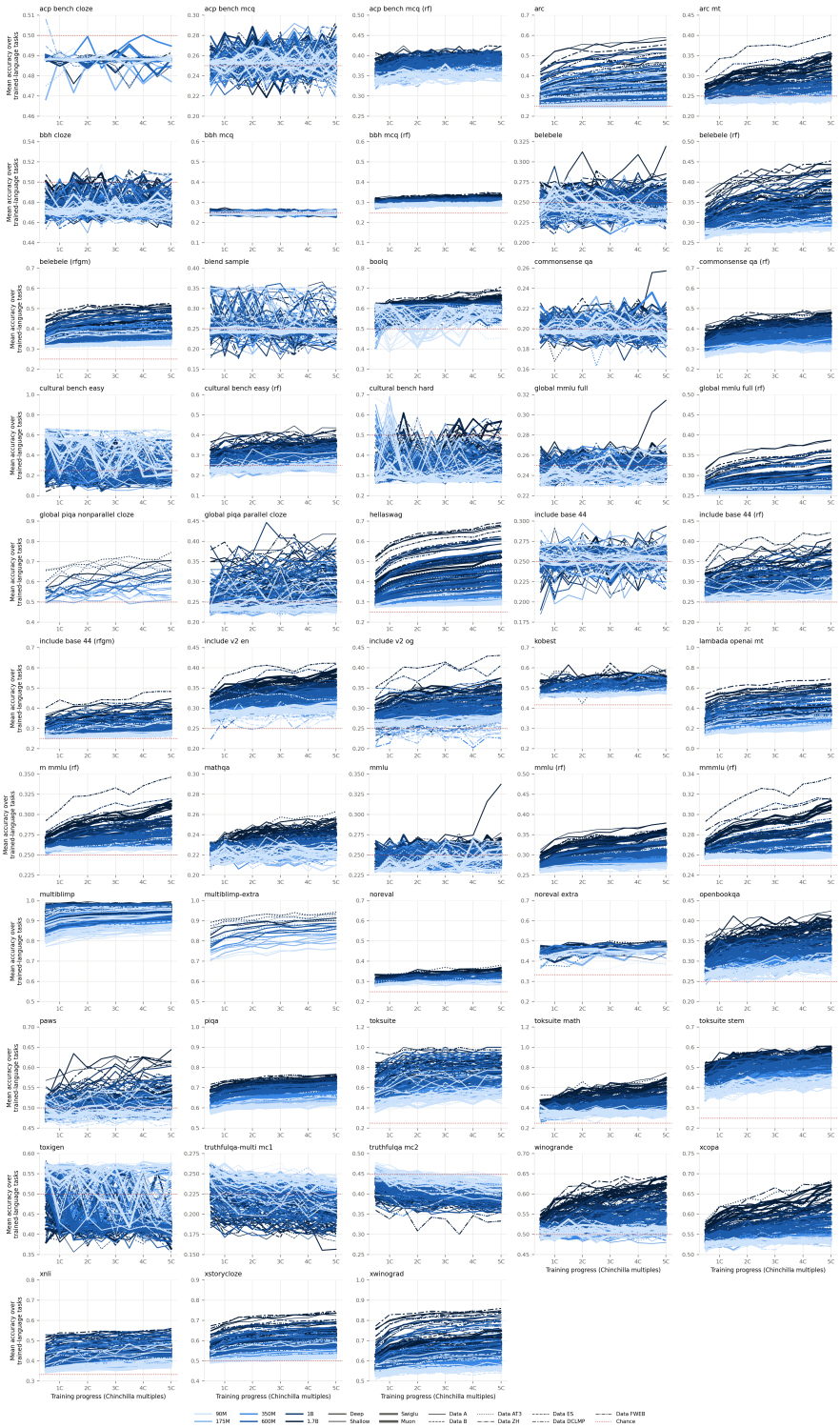


Figure 10: **Along training, the letter-format benchmarks stay at chance while their reformulations improve steadily.** Accuracy at the ten evaluated tenths of each run against Chinchilla multiples of its token budget, averaged over the tasks in the languages the run trains (every seed and data build, 90M–1.7B). Colour indicates the size, width the ladder and style the data build. The dotted red line represents chance. These curves are not filtered by the above-chance threshold.

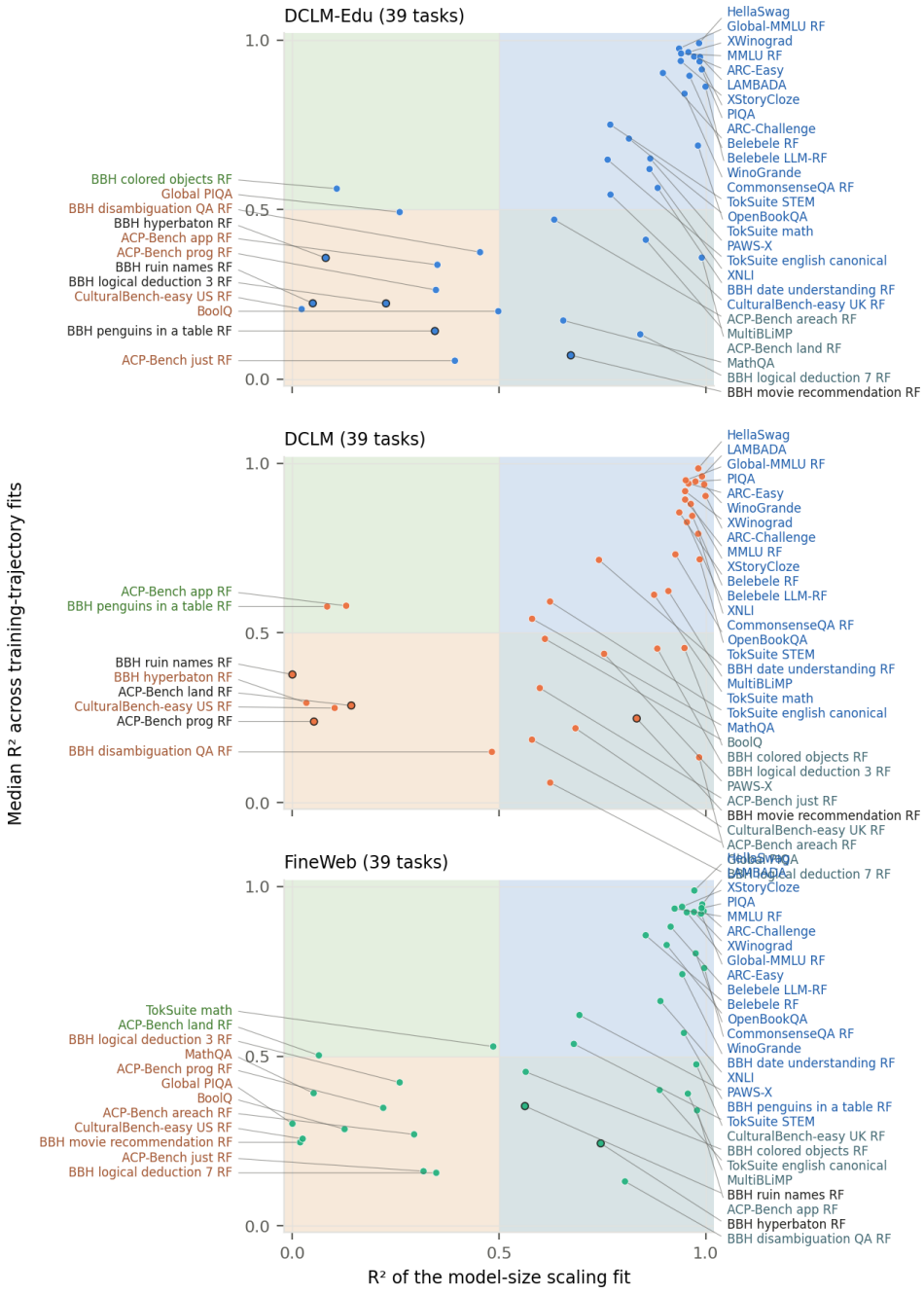


Figure 11: **The commonly used English benchmarks scale predictably.** Each benchmark takes the colour of its regime. ARC-Challenge, ARC-Easy, HellaSwag, LAMBADA, OpenBookQA, Belebele RF, CommonsenseQA RF, Global-MMLU RF, MMLU RF, XNLI, XStoryCloze and XWinograd are predictable across both size and training in all three corpora. Belebele, CommonsenseQA, Global-MMLU, MMLU and TruthfulQA (mc2) never pass the above-chance filter in their original format, and Global PIQA is the one above chance that does not scale.

E Agreement Analyses

This appendix defines ranking agreement and decision accuracy (§E.1), tests their sensitivity to nearties (§E.2) and details the analyses of §5.

E.1 Measuring Ranking Agreement

Settings. Ranking agreement measures whether a set of model variants is ranked in the same order at a cheaper proxy point and at a reference point. A proxy point is a model size and a training progress (N, f). The reference point is the final checkpoint ($f = 1$) either of the same size or of the reference size N_{ref}, 1.7B unless stated otherwise. Table 8 defines the three settings we study. We write RA-size, RA-ckpt and RA-target for ranking agreement in the scale, within-model training and target agreement settings.

Decision accuracy. For a language task b_ℓ, a proxy point (N, f) and a set $\mathcal{P}$ of pairs of model configurations, decision accuracy (DA; Magnusson et al., 2025) is the share of pairs that the proxy orders like the reference. With $s_{ab}(N, f) = \operatorname{sign}\big(Y_{b_\ell}(\theta_a, f) - Y_{b_\ell}(\theta_b, f)\big)$ the order of a pair at a point,

$$\mathrm{DA}(N, f) = \frac{1}{|\mathcal{P}|} \sum_{(a,b)\in\mathcal{P}} \mathbb{1}\big[s_{ab}(N, f) = s_{ab}(N_{\text{ref}}, 1)\big], \quad (2)$$

where the pair is read at size N on the left and at the reference size on the right (N itself for within-model training agreement). A pair tied on both sides counts as agreement and a pair tied on one side as disagreement. DA is 1 when every pairwise preference transfers and 0.5 in expectation for random, tie-free rankings. A cell (b_ℓ, N, f) is reported only if it has at least three pairs. Following Table 8, we call DA-size, DA-ckpt and DA-target for decision accuracy in the scale, within-model training and target agreement settings. Where we restrict to tasks that rank reliably, we keep those whose median DA-size or median DA-ckpt over the proxy points is at least 0.66; a result on these tasks is conditional on that selection.

Comparison axes. When computing RA, we distinguish between two types of model comparisons based on which design choices vary. In *single-axis* comparisons, models differ in only one model design choice while all others are held fixed. For example, we compare deep and shallow architectures, while keeping the language count, language scheme, sampling temperature, optimizer, and activation unchanged. ($\theta' = (D_{\text{deep}}, K, M, T) \to \theta'' = (D_{\text{shallow}}, K, M, T)$). This setting reflects a typical controlled development ablation where we test whether a cheaper proxy leads to the same choice between two alternatives as the reference evaluation. In *multi-axis* comparisons, models may differ in two or more design choices. This tests whether a benchmark preserves rankings across heterogeneous model configurations, similar to how benchmarks are used to rank independently developed models on a leaderboard. We report results for the two settings separately. Figure 3 reports single-axis comparisons; every table is also computed over the multi-axis pairs. Seed replicates are never ranked against each other as design variants, every ranking compares configurations at seed 1904.

Other agreement measures. Over the $n = |\mathcal{P}|$ pairs of model configurations, let C be the number of concordant pairs (ordered the same way and untied on both sides), D the discordant pairs (ordered in opposite ways), T_{both} the pairs tied on both sides, and T_{one} the pairs tied on one side only, so that $n = C + D + T_{\text{both}} + T_{\text{one}}$. Kendall's τ_a and τ_b are

$$\tau_a = \frac{C - D}{n}, \quad \tau_b = \frac{C - D}{\sqrt{(n - T_p)(n - T_r)}},$$

where T_p and T_r are the number of the pairs tied at the proxy and at the reference. Spearman's ρ compares the complete rankings of the V variants, it is the Pearson correlation of their ranks at the proxy and at the reference, with average ranks for ties. Since $\mathrm{DA} = (C + T_{\text{both}})/n$, DA and τ_a differ only in how they treat ties:

$$2\,\mathrm{DA} - 1 = \tau_a + \frac{T_{\text{both}} - T_{\text{one}}}{n}.$$

Over the 6,803 (task, proxy size) values of DA-size, DA correlates with Kendall's τ_b at $r = 0.99$ and with Spearman's ρ at $r = 0.97$ (Figure 12), so we report DA in every analysis.

E.2 Sensitivity of Decision Accuracy

Sensitivity to near-tied reference rankings. Our definition of decision accuracy (DA) requires the ordering of model variants at a proxy point to exactly match their ordering at the reference scale. This strict criterion may overstate decision failures when two variants have nearly identical performance at the reference point: a small numerical

Setting	RA	Proxy	Reference	Question
Scale agreement	RA-size	$(N, f = 1)$	$(N_{\text{ref}}, f = 1)$	How small can the proxy model be?
Within-model training agreement	RA-ckpt	$(N, f < 1)$	$(N, f = 1)$	How early does the ranking stabilize?
Target agreement	RA-target	$(N, f < 1)$	$(N_{\text{ref}}, f = 1)$	How small and early can the proxy be?

Table 8: **Ranking-agreement settings.** The proxy and the reference rank the same model variants but differ in model size, training progress, or both. N is the non-embedding parameter count, N_{ref} the reference size (1.7B, or 3B in Appendix E.6) and f the training progress.

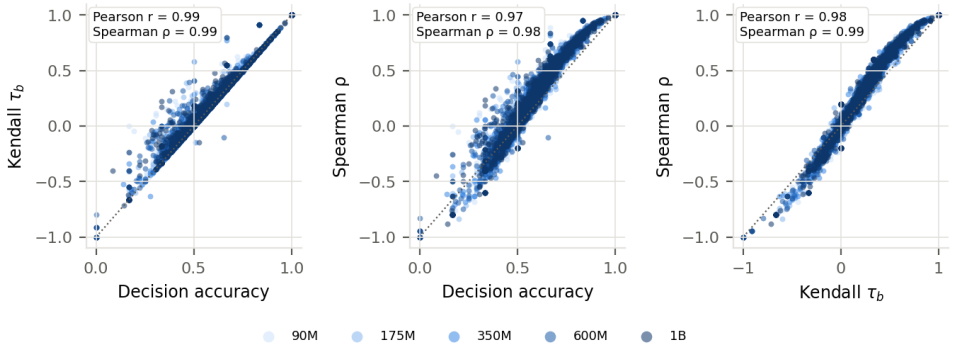


Figure 12: **Decision accuracy, Kendall's τ and Spearman's ρ measure nearly the same ranking agreement.** One point per (benchmark task, proxy size) value of DA-size (6,803 points): the final checkpoint of each proxy against the 1.7B final checkpoint, on the multi-axis pairs of the seed-1904 runs of every data build, over the tasks above chance at the proxy and at 1.7B. Colour marks the proxy size, and each panel gives the Pearson r and Spearman's correlation over the points.

reversal is counted identically to a reversal between substantially different variants. We therefore perform a sensitivity analysis in which reference-scale differences within a tolerance ϵ are treated as effective ties. Specifically, when the proxy and reference rankings disagree but the absolute performance gap between the two variants at the reference point is at most ϵ, we count the comparison as ranking-preserving. We sweep $\epsilon \in \{0, 0.1, 0.25, 0.5, 1, 2\}$ percentage points, with $\epsilon = 0$ corresponding to the original DA definition. The set and denominator of pairwise comparisons remain fixed across thresholds, such that changes in DA arise only from reclassifying near-tied ranking reversals.

Figure 13 shows that increasing the tolerance ϵ raises absolute DA, indicating that some apparent ranking failures are near-ties at the reference scale. At $\epsilon = 0.5$ pp, 755 reversed comparisons are reclassified as correct, increasing to 1,099 at 1 pp and 1,308 at 2 pp (Table 9). However, the DA-size curves remain nearly parallel across thresholds, and the number of reclassified comparisons is similar across proxy sizes. Thus, tolerance mainly shifts DA upward rather than making larger proxy models more predictive of the reference ranking. DA-checkpoint shows the same upward shift while preserving its increase with training progress. Overall, near-ties affect the absolute DA level but do not change the main conclusion: ranking recovery improves with training progress, but not consistently with model size.

ϵ (pp)	Pairs within tolerance at reference	Ranking reversals treated as ties	Already correct
0.1	357	184	173
0.25	982	436	546
0.5	1,949	755	1,194
1.0	3,268	1,099	2,169
2.0	4,301	1,308	2,993

Table 9: **Sensitivity of decision accuracy to the reference-gap tolerance ϵ.** "Within tolerance" counts pairs whose absolute performance difference at the reference scale is at most ϵ. Only reversed pairs are newly reclassified as correct; pairs whose rankings already agree remain unchanged.

Categorizing changes in intervention effects. Decision accuracy measures whether the ordering of model variants is preserved from a proxy point to the reference point, but it does not distinguish between qualitatively different ways in which

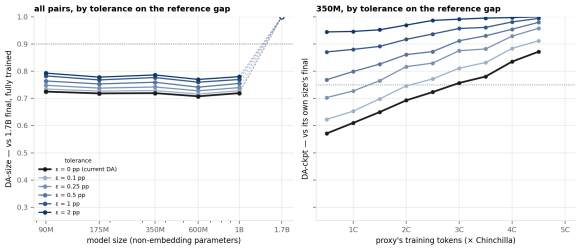


Figure 13: **Sensitivity of decision accuracy to near-tied reference rankings.** Left: DA-size across proxy model sizes. Right: DA-checkpoint across training progress for the 350M model. Increasing the tolerance ϵ raises absolute DA by treating small reference-scale ranking reversals as ties, while largely preserving the shape of the curves. For DA-size, the curves remain nearly parallel across tolerances, indicating that accounting for near-ties does not induce stronger scale convergence. The 1.7B point is the reference scale and therefore reaches perfect agreement by construction.

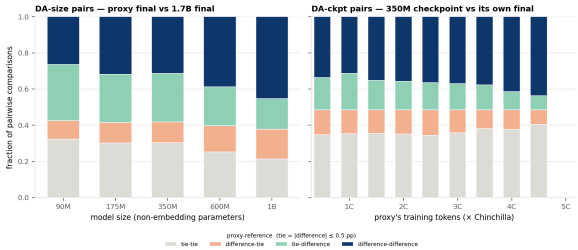


Figure 14: **Evolution of pairwise intervention effects from proxy to reference.** We categorize each pairwise comparison according to whether the absolute performance gap between the two variants is within a tolerance of $\epsilon = 0.5$ percentage points at the proxy and reference points. *Tie–tie* indicates that the variants remain within the tolerance at both points; *difference–tie* that a proxy-scale difference disappears at the reference; *tie–difference* that a difference emerges only at the reference; and *difference–difference* that the variants remain separated at both points. Left: category fractions across proxy model sizes relative to the 1.7B reference model. Right: category fractions across training checkpoints for the 350M model relative to its final checkpoint.

pairwise differences can evolve. In particular, two variants may remain similarly close at both proxy and reference, an apparent proxy-scale difference may disappear at the reference, or a difference may emerge only at larger scale or later in training. To characterize these cases directly, we compare the absolute performance gap between each pair of variants at the proxy and reference points. We use a practical tolerance of $\epsilon = 0.5$ percentage points and treat pairs with an absolute gap of at most ϵ as ties. This yields four categories: *tie–tie*, where the variants remain within the tolerance at both points; *difference–tie*, where a proxy-scale difference washes out at the reference; *tie–difference*, where a difference emerges only at the reference; and *difference–difference*, where the variants remain separated by more than ϵ at both points. We compute these categories using the same pairwise comparisons, filtering, and proxy/reference definitions as in the DA analysis, separately across model sizes and training checkpoints.

Figure 14 shows that larger and later proxies become better at identifying whether an intervention remains consequential at the reference point. For DA-size, *tie–difference* cases decrease with model size while *difference–difference* cases increase, indicating that larger proxies are less likely to miss effects that persist at 1.7B. *Difference–tie* remains comparatively stable, showing that some proxy-scale differences still wash out at the reference. The same pattern appears across training checkpoints, where *tie–difference* decreases and *difference–difference* increases with training progress. Thus, even when exact ranking recovery remains weak, larger models and later checkpoints provide a better signal of whether an intervention will matter at the reference point.

E.3 Decision Accuracy Across Sizes, Answer Formats and Scoring

We measure DA-size, the share of design pairs that a proxy's final checkpoint (90M–1B) orders like the 1.7B reference, on the seed-1904 runs of every language setting, ladder and data build, over every pair of configurations (multi-axis). The tasks are the benchmark tasks above chance at the proxy and at 1.7B with at least three pairs, their bBPB versions (the bits-per-byte of the gold answer on the same items) and the BPB of the 50 trained languages.

Accuracy and bBPB predictably rank the design pairs close to chance. Pooled over pairs, DA-size is 0.58, 0.57, 0.58, 0.58 and 0.58 from 90M to 1B. Per-language BPB ranks them like the reference, at 0.94, 0.94, 0.89, 0.74 and 0.92, and the macro average of BPB over languages at 0.95–0.98.

bBPB predictably ranks design pairs where accuracy is at chance. Where accuracy is at chance at the proxy, bBPB still ranks the pairs like the 1.7B bBPB at 0.60–0.63, above accuracy on the same tasks (0.46–0.50) but far below BPB (Figure 16).

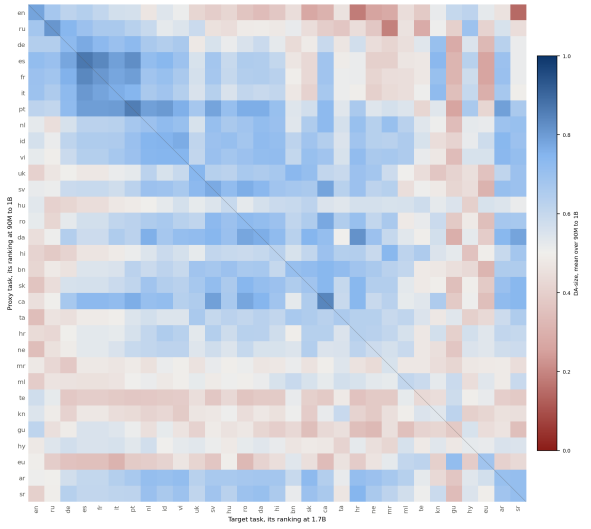


Figure 15: **On HellaSwag, the languages that the $K = 8$ list trains together predict each other's 1.7B ranking, while English does not.** DA-size of the ranking on the HellaSwag task of one language at the proxy (rows) against the 1.7B ranking on another language (columns), averaged over 90M–1B; the diagonal is the task's own DA-size. Single-axis pairs of the seed-1904 runs, at least three per size, each task above chance where it is read. Languages that no list trains (e.g. Basque) are read on the scheme-A runs, which are evaluated on every language. Colour centred at 0.5.

On the tasks that rank reliably, accuracy agrees more with the reference at larger proxies (DA-size from 0.56 at 90M to 0.69 at 1B), while bBPB stays at 0.60–0.65 at every size (Figure 17).

bBPB predicts the reference's accuracy better than accuracy. The comparison above sets bBPB against its own value at 1.7B, an easier target than the reference's accuracy. Against the 1.7B accuracy, on the 360–542 tasks where both readings are defined (multi-axis pairs), the proxy's bBPB agrees more often than the proxy's accuracy: by $+0.10$ at 90M, $+0.06$ at 175M and 350M, $+0.05$ at 600M and $+0.03$ at 1B. bBPB is the better proxy on 57–71% of these tasks (Wilcoxon $p < 10^{-7}$ at every size). The gain is largest at the smallest proxy, where accuracy is closest to chance.

E.4 Decision Accuracy Against the Training Tokens of the Language

We measure the DA-goal of per-language BPB, the share of design pairs that a checkpoint orders like the final 1.7B checkpoint, against the tokens of that language the checkpoint has seen. We use every data build at seed 1904: the proxies 90M–1B at each evaluated tenth of their run. Each of the 50 trained languages gives one task, its BPB, and its pairs are the pairs of configurations that train it, with at least three pairs per language. Each value is the mean over languages at one size and checkpoint.

E.5 Language Transfer Example

We read the ranking of the design variants on the HellaSwag task of one language at each proxy size against the 1.7B ranking on the HellaSwag task of another language, on the single-axis pairs of the seed-1904 runs.

On HellaSwag, languages trained together predict each other's reference rankings. German, Spanish, French and Italian, the four European languages of the $K = 8$ list, predict each other's 1.7B ranking at 0.78 on average, almost as well as each predicts its own (0.79). English predicts the rankings of the other languages at only 0.45 (Figure 15).

E.6 Size Generalisation to 3B

We test whether the conclusions from the 1.7B reference size hold when extending the reference to 3B. We use the six deep configurations trained to 3B at seed 1904 (scheme A at $K \in \{8, 15, 30, 50\}$ and scheme B at $K \in \{8, 15\}$) and the same configurations from 90M to 1.7B. We compare the number of tasks above chance at 1.7B and at 3B, recomputing the threshold on these six configurations, and the DA-size of the proxies against the 3B and the 1.7B final checkpoints on the same decisions. Benchmark accuracy is pooled with bBPB (available for five of the six 3B models, all but $K = 30$), and per-language BPB is pooled separately.

At 3B more tasks pass the above chance threshold but ranking agreement does not improve. On the six configurations, it raises the number of tasks above chance from 612 to 778 of 933 (+27%), mostly the letter-format Belebele, Global-MMLU and INCLUDE tasks (Figure 5). On identical single-axis decisions, DA-size against 3B is 0.49–0.54 for benchmark accuracy and 0.78–0.91 for per-language BPB from every proxy, as against 1.7B, and the 1.7B models themselves reach only 0.51 on accuracy and 0.95 on BPB. The 3B reference also leaves the time course unchanged: every proxy checkpoint reads 0.48–0.58 against either reference, while the reference's own run rises to 0.75–0.77 at 4.5C (Figure 21).

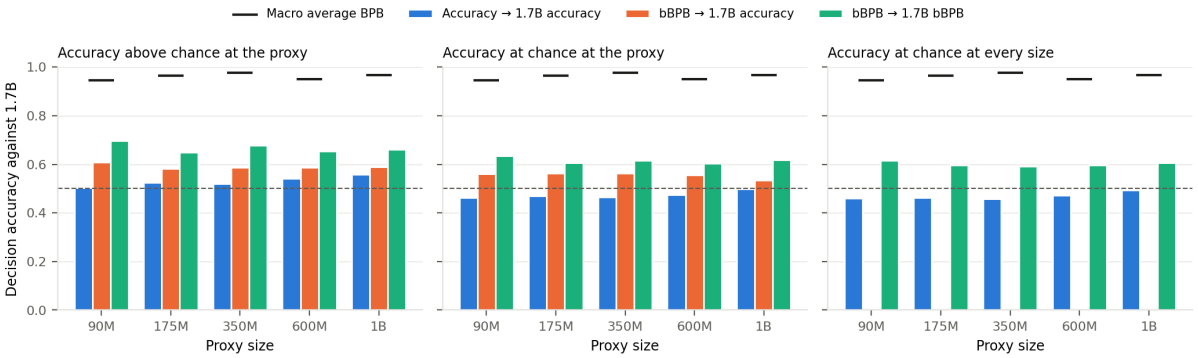


Figure 16: **Where accuracy is at chance at the proxy, bBPB still ranks the design variants like the 1.7B bBPB, though far below per-language BPB.** DA-size against the 1.7B final checkpoint on multi-axis pairs, for the tasks above chance at the proxy (left), at chance at the proxy (middle) and at chance at every size (right). Blue: accuracy against the 1.7B accuracy (no threshold at the proxy). Orange: the proxy's bBPB against the 1.7B accuracy. Green: bBPB against the 1.7B bBPB. Black tick: the macro average of per-language BPB on the same 406 pairs (those with per-item scores). Dashed: 0.5.

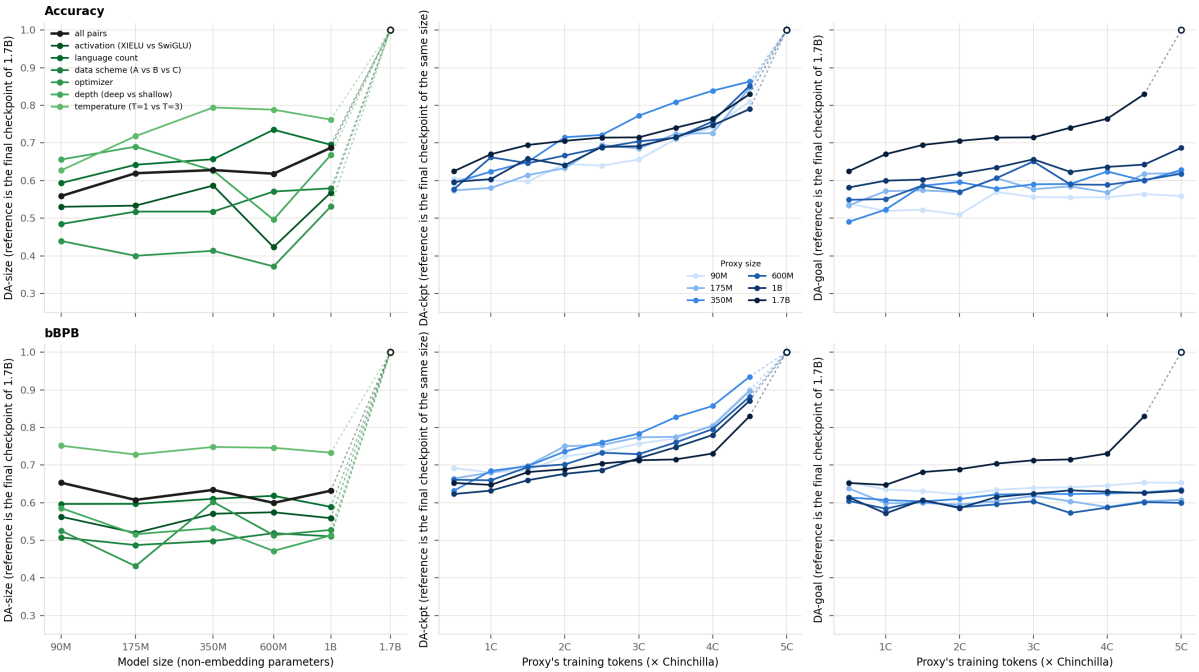


Figure 17: **On accuracy a larger proxy decides more like the 1.7B reference, while on bBPB it does not.** Figure 3 split by scoring, on the same single-axis pairs and task selection: accuracy, including the RF and LLM-RF versions (top), and bBPB, ranked against its own value at the reference (bottom). Left: DA-size, all pairs (black) and one line per design axis. Middle: DA-ckpt. Right: DA-goal. A hollow marker is 1 by construction (a ranking compared with itself).

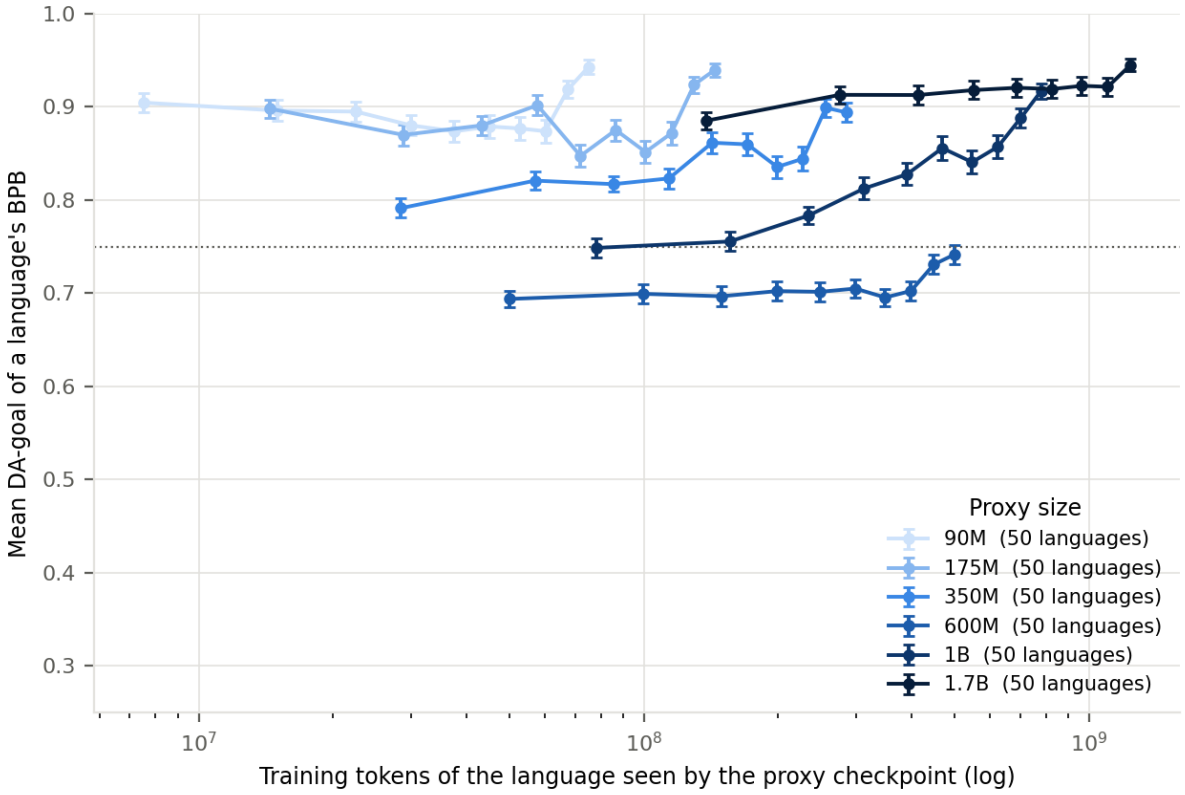


Figure 18: **Mean DA-goal over the BPB of the 50 trained languages against the tokens of the language seen by the checkpoint.**. One line per proxy size and one marker per evaluated tenth, bars give the standard error over languages and the dotted line marks 0.75.

F Anticipation Analyses

This appendix defines the surrogates of decision accuracy (§F.1) and analyses of §6: which SNR definition tracks decision accuracy best (§F.4).

F.1 Surrogates of Decision Accuracy

A *surrogate* is a statistic of a benchmark task that is computed at the proxy size alone, without the reference, and is meant to indicate whether the proxy's ranking will agree with the reference (§6). This appendix defines the surrogates of Figure 4.

Notation. A task is scored on the V model variants of one proxy size. Variant a has a final score x_a and a *noise window* W_a of five checkpoints in the last 20% of its run (80, 85, 90, 95 and 100%), with mean $\bar{w}_a$ and population standard deviation s_a. Scores are oriented so that higher is better, so BPB and bBPB enter with a flipped sign.

F.2 Signal-to-Noise Ratio

The signal-to-noise ratio (SNR) summarizes a benchmark's reliability as the ratio between signal, how well a benchmark separates models of similar compute, and noise, the variability in scores across training checkpoints or stochastic runs. We define it as

$$\mathrm{SNR} = \frac{\mathrm{Signal}}{\mathrm{Noise}}.$$

A higher SNR means score differences across models are large relative to the benchmark's intrinsic variability. The framework is not tied to a particular signal or noise definition, so we evaluate several candidates and select the pair with the strongest empirical correlation to DA.

F.2.1 Signal candidates

Figure 4 uses three candidate signal definitions to capture a different notion of how spread out model scores are. The **relative standard deviation** is the population standard deviation of the final scores relative to their mean,

$$S_{\mathrm{rel}} = \frac{1}{\bar{x}} \sqrt{\frac{1}{V} \sum_{a=1}^{V} (x_a - \bar{x})^2}.$$

The **distance standard deviation** (`dist_std`) is the population standard deviation of the P =

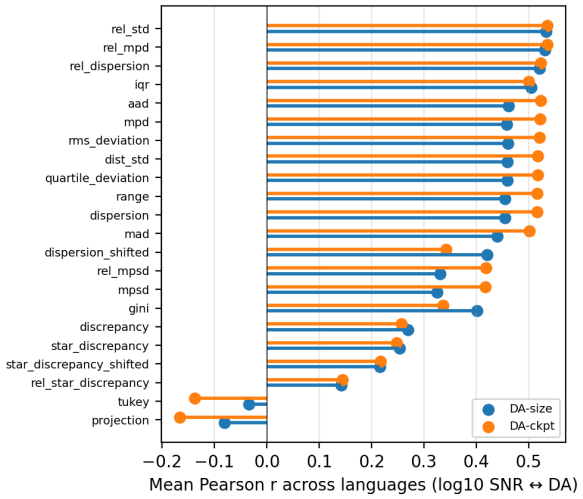


Figure 19: **The relative-spread SNR definitions track decision accuracy best, as a block.** For each of the 22 SNR definitions, the mean over languages of the Pearson r between $\log_{10}$ SNR and decision accuracy across the tasks of one language, for DA-size (blue) and DA-ckpt (orange). The definitions are ordered by their DA-size correlation.

$V(V-1)/2$ pairwise distances $d_{ab} = |x_a - x_b|$, $a < b$, with mean $\bar{d}$:

$$S_{\text{dist}} = \sqrt{\frac{1}{P}\sum_{a<b}\left(d_{ab} - \bar{d}\right)^2}.$$

The **discrepancy** is the centred L_2 discrepancy of the final scores read as values in $[0, 1]$, which measures how far they are from evenly spread. In the squared form computed by SciPy's `qmc.discrepancy`, with $u_a = |x_a - \frac{1}{2}|$,

It is defined for accuracy only. The *signal alone* in Figure 4 is S_{rel}.

F.2.2 Noise candidates

Seed noise. Standard deviation of the final checkpoint across training runs with different random seeds.

$$\text{Seed noise}(M) = \sigma(M),$$
$$M_i = \tfrac{1}{n}\textstyle\sum_{j=T-n+1}^{T} U(t_j)$$

Total variation. Average change in metric score across training checkpoints minus an improvement term.

$$\text{Total Variation} =$$
$$\tfrac{1}{T}\textstyle\sum_{t=1}^{T}|U(t) - U(t-1)| - \tfrac{1}{T}(U(T) - U(0))$$

Checkpoint-to-checkpoint noise. Proposed by Heineman et al. (2025) as a proxy when the first three are too expensive to estimate at large scale and depends only on the final n training checkpoints:

$$\text{Checkpoint noise} = \sigma\big(\{U(t_j)\}_{j=T-k+1}^{T}\big).$$

On our ladder both checkpoint noise, over the last five evaluated checkpoints of each run, and seed noise, over the cells trained with three seeds, are measurable, and §6 reports every intervention's effect against the seed noise.

Benchmark noise. Average relative standard deviation of scores across $k = 5$ disjoint folds of the evaluation set:

$$\text{Noise}(M) = \frac{1}{|M|}\sum_{m\in M}\frac{\sqrt{\frac{1}{k}\sum_{i=1}^{k}(m_i - \bar{m})^2}}{\bar{m}}$$

where $m_1, \ldots, m_k$ are the per-fold scores of model m. The fold scores are obtained by partitioning the stored per-question outputs of a single evaluation run, so no inference is repeated and no intermediate checkpoints are needed.

F.3 Other Surrogates

Scaling-fit R^2. The median R^2 of the log-linear fits of a task's final score against $\log_{10} N$ (Appendix D.4), using only the sizes up to the proxy. A fit needs three sizes, so this surrogate exists from 350M.

Margin above chance. The task's mean score at the proxy size minus its chance level c_b (Appendix D.1),

$$M = \bar{x} - c_b.$$

It is defined for accuracy only.

Early-checkpoint agreement. The DA-ckpt of the proxy's checkpoint at 10% of its run against its own final checkpoint, $\text{DA}(N, 0.1)$ in the within-model training setting (Table 8).

F.4 SNR as a Surrogate for Decision Accuracy

We compute 22 SNR definitions: each divides a spread of the final scores of the design variants at a size by the checkpoint noise of the last 20% of the runs, relative to the mean score of that window. For each definition and language, we correlate $\log_{10}$ SNR with decision accuracy (Pearson r) across the tasks of that language, over every pair of configurations (multi-axis) and pooling the proxy sizes

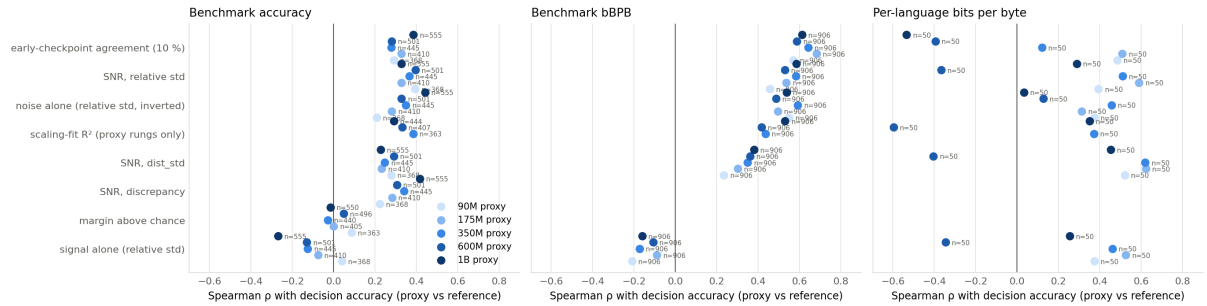


Figure 20: **bBPB raises the level of the correlations in Figure 4, on accuracy alone the relative-standard-deviation SNR leads or ties.** Spearman ρ between each surrogate and DA-size against 1.7B at each proxy size, on the benchmark tasks above chance scored by accuracy (left), their bBPB versions (middle) and per-language BPB (right). The number next to each marker is the number of tasks. The discrepancy and the margin above chance exist for accuracy only, the scaling-fit R^2 needs three sizes and starts at 350M.

90M–1B. The tasks are those above chance at the proxy (and, for DA-size, at 1.7B), plus bBPB and per-language BPB, which have no chance level. We average the correlation over the languages with at least three tasks, for DA-size and for DA-ckpt (the nine earlier tenths of a proxy's run against its own final checkpoint).

The relative-spread group is the one that correlates better with decision accuracy. (Figure 19).

bBPB drives the level of the correlations. Split by scoring, bBPB raises both the level of the correlations in Figure 4 and the lead of early-checkpoint agreement: on bBPB, early-checkpoint agreement reaches $\rho = 0.57$–0.68 and the relative-standard-deviation SNR 0.46–0.58, against 0.28–0.39 and 0.33–0.40 on accuracy (Figure 20). On per-language BPB, five of the six statistics turn negative at 600M, the size whose BPB ranks the variants least like the reference. On single-axis pairs alone, all correlations are weaker, and the relative-standard-deviation SNR leads at every proxy size.

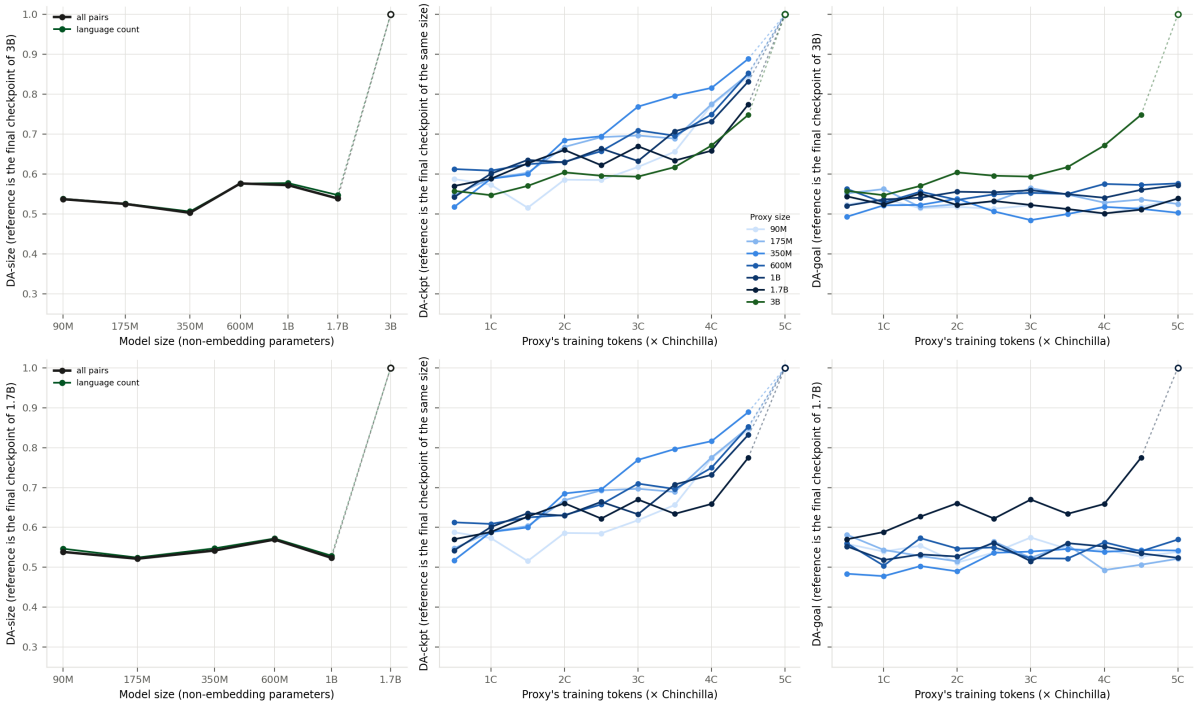


Figure 21: **A 3B reference ranks the proxies no better than the 1.7B one.** Figure 3 on the six configurations trained at 3B, against the 3B (top) and the 1.7B (bottom) final checkpoints. Both rows read the nine single-axis pairs of these configurations and the 553 tasks reliable under both references (394 bBPB), so they differ in the reference alone. **(a)** DA-size; only the number of languages has three or more pairs. **(b)** DA-ckpt, which does not depend on the reference. **(c)** DA-goal.